



RKDF UNIVERSITY RANCHI

**DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING
(Diploma EEE)**



**DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING
(Diploma EEE)**



RKDF UNIVERSITY RANCHI

DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING (Diploma EEE)

Diploma Engineering Semester-I (Common to all Branches)

Credit System and Marks Distribution (2023 batch):-

Semester-I												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	BSC	DE101	Applied Chemistry	3	0	0	3	30	70	21	100	35
2	BSC	DE102	Mathematics - I	3	0	0	3	30	70	21	100	35
3	Huminties & Social Sc. Courses	DE103	Communication Skills in English	2	0	0	2	30	70	21	100	35
4	BSC	DE104	Applied Physics	3	0	0	3	30	70	21	100	35
5	ESC	DE105	Fundamentals of Computer	2	0	0	2	30	70	21	100	35
6	ESC	DE106	Engineering Graphics	2	0	0	2	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	BSC	DE151	Applied Chemistry Lab	0	0	2	1	30	20		50	25
2	Huminties & Social Sc. Course	DE153	Communication Skills in English	0	0	2	1	30	20		50	25
3	BSC	DE154	Applied Physics Lab	0	0	2	1	30	20		50	25
4	ESC	DE155	Fundamentals of Computer Lab	0	0	2	1	30	20		50	25
5	ESC	DE156	Engineering Graphics Lab	0	0	3	1.5	30	20		50	25
6	ESC	DE157	Workshop / Manufacturing Practices Lab	0	0	3	1.5	30	20		50	25
TOTAL							22					



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Credit System and Marks Distribution (2023 batch):-

Semester-II												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	BSC	DE201	Mathematics – II	3	0	0	3	30	70	21	100	35
2	BSC	DE202	Applied Physics – II	3	0	0	3	30	70	21	100	35
3	ESC	DE203	Fundamentals of Electrical & Electronics Engineering	2	0	0	2	30	70	21	100	35
4	ESC	DE204	Engineering Mechanics	2	0	0	2	30	70	21	100	35
5	Humanities & Social Sc. Course	DE205	Communication Skills in EnglishII	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	BSC	DE252	Applied Physics –II Lab	0	0	2	1				50	25
2	ESC	DE253	Fundamentals of Electrical & Electronics Engineering Lab	0	0	2	1				50	25
3	ESC	DE254	Engineering Mechanics Lab	0	0	2	1				50	25
TOTAL							16					



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Credit System and Marks Distribution (2023 batch): -

Semester-III												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	BSC	DEEP301	Mathematics – III	3	0	0	3	30	70	21	100	35
2	ESC	DEEP302	Object Oriented Programming with C++	2	0	0	2	30	70	21	100	35
3	PCC	DEEP303	Network Theory	3	0	0	3	30	70	21	100	35
4	PCC	DEEP304	Electrical and Electronic Measurement and Instrumentation	3	0	0	3	30	70	21	100	35
5	PCC	DEEP305	Principal of Digital Electronics	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	PCC	DEEP351	Network Theory Lab	0	0	2	1				50	25
2	PCC	DEEP352	Electrical and Electronic Measurement and Instrumentation Lab	0	0	2	1				50	25
3	PCC	DEEP353	Principal of Digital Electronics Lab	0	0	2	1				50	25
4	ESC	DEEP354	Object Oriented Programming with C++ Lab	0	0	2	1				50	25
6	IN		Summer Internship (4 weeks) after 2 nd sem	0	0	4	2				50	25
TOTAL							20					



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Credit System and Marks Distribution:-

Semester-IV												
SL. No.	Category	SubjectCode	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	PCC	DEEP401	Power System-I	3	0	0	3	30	70	21	100	35
2	PCC	DEEP402	Electrical Machine-I	3	0	0	3	30	70	21	100	35
3	PCC	DEEP403	Applied Electronics	3	0	0	3	30	70	21	100	35
4	PCC	DEEP404	Electrical Estimation and Costing	3	0	0	3	30	70	21	100	35
5	PEC	DEEP(E)	Program Elective-I	3	0	0	3	30	70	21	100	35
6	PEC	DEEP(E)	Program Elective-II	3	0	0	3	30	70	21	100	35
Program Elective-I (Choose any one)												
1	PEC	DEEP405(E)	Signals and Systems	3	0	0	3	30	70	21	100	35
2	PEC	DEEP406(E)	Communication technology	3	0	0	3	30	70	21	100	35
3	PEC	DEEP407(E)	Illumination practices	3	0	0	3	30	70	21	100	35
Program Elective-II (Choose any one)												
1	PEC	DEEP408(E)	Utilization of Electrical Energy	3	0	0	3	30	70	21	100	35
2	PEC	DEEP409(E)	Electric Traction	3	0	0	3	30	70	21	100	35
3	PEC	DEEP410(E)	Electrical Testing and commissioning	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	PCC	DEEP451	Power System -I Lab	0	0	2	1				50	25
2	PCC	DEEP452	Electrical Machine I lab	0	0	2	1				50	25
3	PROJ		Minor Project	0	0	4	2				50	25
4			Essence of Indian Knowledge & Tradition	0	0	0	0				50	25
TOTAL							22					



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Credit System and Marks Distribution:-

Semester-V												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	PCC	DEEP501	Power System -II	3	0	0	3	30	70	21	100	35
2	PCC	DEEP502	Electrical Machine -II	3	0	0	3	30	70	21	100	35
3	PCC	DEEP503	Microprocessor & Microcontroller	2	0	0	2	30	70	21	100	35
4	PEC	DEEP(E)	Program Elective-III	3	0	0	3	30	70	21	100	35
5	PEC	DEEP(E)	Program Elective-IV	3	0	0	3	30	70	21	100	35
6	OEC	DEEO(E)	Open Elective-I	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	PCC	DEEP551	Power system II Lab	0	0	2	1				50	25
2	PCC	DEEP52	Electrical Machine -II Lab	0	0	2	1				50	25
	PCC	DEEP553	Microprocessor & Microcontroller Lab	0	0	2	1					
3	IN	DE554	Summer Internship-II (after 4 th sem)	0	0	0	3				50	25
TOTAL							23					



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Elective Papers for semester-V

Program Elective-III (Choose any one)

Program Elective-III (Choose any one)												
SL No	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	PEC	DEEP504(E)	Switch gear and Protection	3	0	0	3	30	70	21	100	35
2	PEC	DEEP505(E)	Wind and Solar energy System	3	0	0	3	30	70	21	100	35
3	PEC	DEEP506(E)	Electric Drives	3	0	0	3	30	70	21	100	35

Program Elective-IV (Choose any one)

1	PEC	DEEP507(E)	Industrial Instrumentation	3	0	0	3	30	70	21	100	35
2	PEC	DEEP508 (E)	Solar Technology	3	0	0	3	30	70	21	100	35
3	PEC	DEEP509(E)	biomedical	3	0	0	3	30	70	21	100	35

Open Elective-I (Choose any one)

1	OEC	DEEO510(E)	Solid Waste Management	3	0	0	3	30	70	21	100	35
2	OEC	DEEO511(E)	Power Plant Engineering	3	0	0	3	30	70	21	100	35
3	OEC	DEEO512(E)	Web Technology	3	0	0	3	30	70	21	100	35



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Credit System and Marks Distribution: -

Semester-VI												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	PCC	DEEP601	Power Electronics	3	0	0	3	30	70	21	100	35
2	PCC	DEEP602	Control Engineering	3	0		3	30	70	21	100	35
3	Huminitie s & social Sc. Course	DEEH603	Entrepreneurship	3	0	0	3	30	70	21	100	35
4	Mandator y Course	DEEM604	Environmental Science	0	0	0	0	30	70	21	100	35
5	OEC	DEEO(E)	Open Elective-II	3	0	0	3	30	70	21	100	35
6	OEC	DEEO(E)	Open Elective-III	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	PCC	DEEP651	Power Electronics Lab	0	0	2	1				50	25
2	PCC	DEEP652	Control Engineering Lab	0	0	2	1				50	25
3	PROJ	DE654	Major Project	0	0	8	4				50	25
4	SEM	DE655	Seminar	0	0	2	1				50	25
TOTAL							22					



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Elective Papers for semester-VI

Open Elective-II (Choose any one)

SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	OEC	DEEO604(E)	PLC & SCADA	3	0	0	3	30	70	21	100	35
2	OEC	DEEO605(E)	Electrical hybrid Vehicle	3	0	0	3	30	70	21	100	35
3	OEC	DEEO606(E)	Biomass and Micro-hydro Power Plants	3	0	0	3	30	70	21	100	35
Open Elective-III (Choose any one)												
1	OEC	DEEO607(E)	Robotics	3	0	0	3	30	70	21	100	35
2	OEC	DEEO608(E)	Green Building & Energy Conservation	3	0	0	3	30	70	21	100	35
3	OEC	DEEO609(E)	Introduction to Artificial Intelligence	3	0	0	3	30	70	21	100	35



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Diploma Engineering Semester-I

(Common to all Branches)

Course Code	DE101
Course Title	Applied Chemistry
Credits	3 (L: 3 T: 0 P: 0)

Course Objectives:

1. The students will acquire a foundation in chemistry of sufficient breadth and depth to enable them to understand and critically interpret the primary chemical literature.
2. The students will develop the ability to effectively communicate scientific information and research results in written and oral formats
3. The students will learn professionalism, including the ability to work in teams and apply basic ethical principles.

UNIT 1

Atomic Structure and Chemical Bonding:

Atomic Structure: Definition of atom, Fundamental particles of atom –electron, neutron, protons, Definition of Atomic no, Atomic mass no Isotopes & Isobars, & their distinction with suitable examples, Bohr's and Rutherford's theory; Definition and Shape of the orbital & distinction between orbits and orbitals, Hund's Rule, filling up the orbital's by Aufbau's principle (till Atomic no. 30) Chemical bonding: Cause of chemical bonding, types of bonds: electrovalent, covalent and coordinate bonds, formation of electrovalent & covalent compounds

Eg. NaCl, CaCl₂, CO₂, Cl₂, NH₃, C₂H₄, N₂, C₂H₂, etc., coordination bond in NH₄⁺, and anomalous properties of NH₃, HO₂ due to hydrogen bonding, and metallic bonding, Octet rule, Duplet rule.

UNIT 2

Water and Corrosion:

Water: Characteristics, Sources, Impurities, Hard & Soft Water, Causes of Hardness, Types of Hardness, Degree of Hardness, Boiler and Steam Generation, Scale & Sludge Formation – Causes, Disadvantage, Softening Methods such as Boiling, Clark's, Soda Ash, Lime Soda, Zeolite & Ion Exchange Methods with Principle Chemical Reactions. Plumbo solvency & its Removal. Numerical Problems. Corrosion: Definition of Corrosion, Types of Corrosion (Dry and Wet chemical Corrosion) and their mechanism.



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Protection of metal from corrosion (Corrosion Control). Application of Protective Coatings like metal coating such as Galvanizing, Tinning, Metal Spraying, Sherardizing, Electroplating and Metal Cladding.

UNIT 3

Electrochemistry and Electrochemical Cells:

Electrochemistry: Definition of terms: Conductors, Insulators, Dielectrics, Electrolyte, Non-Electrolyte, Electrolysis, Electrolytic Cell with suitable examples, Electrolytic dissociation, Arrhenius Theory of Ionization, Degree of Ionization & factors affecting degree of ionization. Redox reactions. Faradays laws of electrolysis and simple numerical problems. Electrochemical cells: Concept of electrode potential such as reduction potential & oxidation potential. Electrochemical Series, Electrolysis of CuSO_4 Solution by using Cu Electrode & Platinum Electrode, Electrochemical Cells & Batteries, Definition, types such as Primary & Secondary Cells & their examples.

UNIT 4

Lubricants and Fuels:

Lubricants: Definition, Classification with examples. Functions of lubricant, Lubrication -Mechanism of Lubrication (Fluid Film, Boundary and Extreme Pressure). Physical Characteristics of Lubricants Such as Viscosity, Viscosity Index, Oiliness, Volatility, Flash & Fire Point, and Cloud & Pour Point, Chemical Characteristics such as Acid Value or Neutralization Number, Emulsification, Saponification Value, Selection of Lubricants, Characteristics of Transformer oil Fuels: Definition and combustion of fuel, classification of fuels, calorific values (HCV and LCV), calculation of HCV and LCV using Dulong's formula. Proximate analysis of coal solid fuel petrol and diesel - fuel rating (octane and cetane numbers), Chemical composition, calorific values and applications of LPG, CNG, water gas, coal gas, producer gas and biogas.

UNIT 5

Metals and Non - Metallic Materials:

Metals: Occurrence of Metal such as Iron, Aluminum, Chromium, Nickel, Tin, their properties Definition of Metallurgy Mineral, Ore Gangue Flux & Slag, stages of Extraction of metal from Its Ores in detail. Alloys: definition of alloy, purposes of making alloy. preparation methods, General Principal of metallurgy, minerals/ ores, ore dressing, roasting, smelting, bessemerization, fluxes, purification. Explanation of alloying purposes, methods of preparation, composition and uses of alloy like brass, bronze, duralumin, German silver, gun metal, solder, stainless steel, casting and bearing alloy. Nonmetallic materials: Definition of Polymers, formation of Polymers by Addition & condensation polymerization. Properties and uses of PVC, polyethene, polystyrene, polyamides, polyesters, Bakelite. Synthetic fibers – nylon, rayon, decron, and polyesters. Natural Rubber rubber its processing and drawbacks, vulcanization



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of rubber with chemical reaction, synthetic rubber, definition & distinction between natural & synthetic rubber Thermal insulating material: definition & characteristics of thermal insulator. Preparation, properties & application of thermocol & glass wool Properties & application of Asbestos cork

Text Books:

1. Text Book of Chemistry for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi, 2017-18. TEX

Reference Books

1. Engineering Chemistry Jain & Jain, Dhanpat Rai and Sons; New Delhi, 2015
2. Engineering Chemistry S. S. Dara & Dr.S.S.Umare, S. Chand Publication; New Delhi, 2015.
3. Industrial Chemistry B. K. Sharma, Goel Publication
4. Engineering Chemistry By O P Agarwal (Khanna Publication)
5. Environmental Chemistry & Pollution Control S. S. Dara, S. Chand Publication
6. Chemistry for Engineers Agnihotri, Rajesh, Wiley India Pvt. Ltd., 2014.
7. Engineering Chemistry Rao and Agarwal
8. Engineering Chemistry P.C. Jain
9. Applied Chemistry H.N. Sahni
10. Polytechnic Chemistry Vedprakash Mehta, Jain brothers.
11. Engineering Chemistry Uppal

Course Outcomes:

At the end of the course student will be able to

1. Understand and assess the suitability of water source for domestic and industrial application, effluents and minimize water pollution.
2. Qualitatively analyze the engineering materials and understand their properties and applications.
3. Choose fuel and lubricants suitable for economical industrial processing to obtain eco-friendly finished products.
4. a) Ascertain construction, mechanism efficiency of electrochemical cells, solar cell fuel cells
b) Understand corrosion and develop economical prevention techniques.



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Course Code	DE151
Course Title	Applied Chemistry Lab
Credits	1 (L: 0 T: 0 P: 2)



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(Diploma EEE)

Diploma Engineering Semester-I

(Common to all Branches)

Course Objectives:

There are numerous number of materials used in fabricating and manufacturing devices for the com- fort of life. The selection, characterization and suitability assessment of natural raw materials essentially requires principles and concepts of Applied Chemistry for technicians. The course aims to supplement the factual knowledge gained in the lectures by first hand manipulation of processes and apparatus. This will develop scientific temper and help to apply the basic concepts and principles in solving engineering problems.

List of Experiments:

1. Determination of Total hardness by EDTA method.
2. Determination of Total hardness by Clarke's method.
3. Determination of Flash & Fire Points by Pensky Marten Apparatus.
4. Determination of Flash & Fire Points by by Abel's Apparatus.
5. Determination of Viscosity and Viscosity index by Redwood viscometer No.1.
6. Determination of Viscosity and Viscosity index by Redwood viscometer No.2.
7. Determination of percentage of Copper in Brass by Iodometric Titration.
8. To prepare a solution of N/20 sodium carbonates and find the strength of HCl using N/20 sodium carbonate solution.

Reference Books:

1. Applied Chemistry Laboratory Practices Vol. I and Vol. II, Dr. G. H. Hugar & Prof A. N. Pathak ,NITTTR, Chandigarh, Publications, 2013-14. LAB
2. Dr. G. H. Hugar and Prof A. N. Pathak, Applied Chemistry Laboratory Practices, Vol. I and Vol. II,NITTTR, Chandigarh, Publications, 2013-14.
3. Agnihotri, Rajesh, Chemistry for Engineers, Wiley India Pvt.Ltd., 2014.
4. Jain & Jain, Engineering Chemistry, Dhanpat Rai and Sons; New Delhi, 2015.



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Learning Outcomes:

At the end of the course student will be able to

- To express quantitative measurements accurately.
- To practice and adapt good measuring techniques.
- To use various apparatus for precise measurements.
- To understand and differentiate different methods of quantitative analysis.
- To know and understand principles of quantitative analysis using instruments.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Engineering Semester-I

(Common to all Branches)

Course Code	DE104
Course Title	Applied physics-I
Credits	3 (L: 3 T: 0 P: 0)

Course Objectives:

Course objectives- study of applied physics aims to give an understanding of physical world by observation and predictions. The course help diploma engineers to apply the basic concepts and principles to solve engineering problems and to understand different technology-based application.

Unit -1

Physical world, Units and Measurements: Physical quantities; fundamental and derived, Units and systems of units (FPS, CGS and SI units), Dimensions and dimensional formulae of physical quantities, Principle of homogeneity of dimensions, Dimensional equations and their applications (conversion from one system of units to other, checking of dimensional equations and derivation of simple equations), Limitations of dimensional analysis. Measurements: Need, measuring instruments, least count, types of measurement (direct, indirect), Errors in measurements (systematic and random), absolute error, relative error, error propagation, error estimation and significant figures.

Unit-2

Force and Motion: Scalar and Vector quantities – examples, representation of vector, types of vectors. Addition and Subtraction of Vectors, Triangle and Parallelogram law (Statement only), Scalar and Vector Product, Resolution of a Vector and its application to inclined plane and lawn roller. Force, Momentum, Statement and derivation of conservation of linear momentum, its applications such as recoil of gun, rockets, Impulse and its applications. Circular motion, definition of angular displacement, angular velocity, angular acceleration, frequency, time period, Relation between linear and angular velocity, linear acceleration and angular acceleration (related numerical), Centripetal and Centrifugal forces with live examples, Expression and applications such as banking of roads and bending of cyclist.



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Unit-3

Work, Power and Energy: Work Concept and units, examples of zero work, positive work and negative work Friction: concept, types, laws of limiting friction, coefficient of friction, reducing friction and its engineering applications, Work done in moving an object on horizontal and inclined plane for rough and plane surfaces and related applications. Energy and its units, kinetic energy, gravitational potential energy with examples and derivations, mechanical energy, conservation of mechanical energy for freely falling bodies, transformation of energy (examples). Power and its units, power and work relationship, calculation of power (numerical problems)

Unit-4

Rotational Motion: Translational and rotational motions with examples, Definition of torque and angular momentum and their examples, Conservation of angular momentum (quantitative) and its applications. Moment of inertia and its physical significance, radius of gyration for rigid body, theorems of parallel and perpendicular axes (statements only), Moment of inertia of rod, disc, ring and sphere (hollow and solid); (Formulae only).

Unit-5

Properties of Matter: Elasticity: definition of stress and strain, moduli of elasticity, Hooke's law, significance of stress-strain curve. Pressure: definition, units, atmospheric pressure, gauge pressure, absolute pressure, Fortin's Barometer and its applications. Surface tension: concept, units, cohesive and adhesive forces, angle of contact, Ascent Formula (No derivation), applications of surface tension, effect of temperature and impurity on surface tension. Viscosity and coefficient of viscosity: Terminal velocity, Stoke's law and effect of temperature on viscosity, application in hydraulic systems. Hydrodynamics: Fluid motion, stream line and turbulent flow, Reynold's number Equation of Continuity, Bernoulli's Theorem (only formula and numerical) and its applications.

Heat and Thermometry: Concept of heat and temperature, modes of heat transfer (conduction, convection and radiation with examples), specific heats, scales of temperature and their relationship, Types of Thermometers (Mercury thermometer, Bimetallic thermometer, Platinum resistance thermometer, Pyrometer) and their uses. Expansion of solids, liquids and gases, coefficient of linear, surface and cubical expansions and relation amongst them, Co-efficient of thermal conductivity, engineering applications.



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Text Books:

1. Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi

Reference Books:

1. Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi.
2. Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi
3. Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi
4. Engineering Physics by DK Bhattacharya & Poonam Tandan; Oxford University Press, New Delhi

Course Outcomes:

After undergoing this subject, the students will be able to-

1. Identify physical quantity, select their units for engineering solutions and make measurement with accuracy.
2. Describe the different forms of energy, methods of transfer of energy.
3. Represent the physical quantity as scalar and vectors to solve real life relevant problems.
4. Analyze the type of motion and apply the formulation to understand banking of road and conservation of momentum, recoil of gun etc.



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Diploma Engineering Semester-I (Common to all Branches)

Course Code	DE154
Course Title	Applied Physics-I Lab
Credits	1 (L: 0 T: 0 P: 2)

Couse objectives-

The course aims to supplement the factual knowledge gained in the lecture by first hand manipulation of apparatus. This will develop scientific temper and help student to apply the basic concepts in solving engineering and technology-based problems. In addition, students geta confidence in handling equal and thus learn skill of measurements.

List of Practical:

- 1) To measure the length, radius of a given cylinder, a test tube using vernier caliper and find the volume of each object.
- 2) To determine the diameter of a wire, a solid ball using screw gauge.
- 3) To Verify the Ohm's law
- 4) To verify the Kirchhoff's law (kvl ans kcl)
- 5) To find the surface tension of a liquid by capillary rise method.
- 6) To determine the viscosity of a given liquid (Glycerin) by Stoke's law.
- 7) To verify the law of conservation of mechanical energy.
- 8) To measure the room temperature of hot bath using mercury Thermometer and convert it into different scale.
- 9) To determine the radius of curvature of concave and convex mirror using spherometer.
- 10) Determine the force constant using Hook's law.



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Reference Books:

1. Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi.
2. Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi
3. Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi
4. Engineering Physics by DK Bhattacharya & Poonam Tandan; Oxford University Press, New Delhi

Course Outcomes:

After undergoing this lab work, students will be able to -

1. Select the right kind of measuring tools (meter scale, screw gauge, Vernier calipers etc.)
2. Describe and verify the Hook's law and determine the force constant of a spring body.
3. Understand the Stoke's law and determine the viscosity of a given liquid.
4. Understand the use of thermometer to measure the temperature under different conditions and scales.
5. Understand the current and voltage relationship.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Engineering Semester-I

(Common to all Branches)

Course Code	DE103
Course Title	Communication skills in English
Credits	2 (L: 2 T: 0 P: 0)

Course Objectives:

Communication skills play an important role in career development. This course aims at introducing basic concepts of communication skills with an emphasis on developing personality of the students. Thus, the main objectives of this course are:

To develop confidence in speaking English with correct pronunciation.

To develop communication skills of the students i.e. listening, speaking, reading and writing skills

To introduce the need for personality development- Focus will be on developing certain qualities which will aid students in handling personal and career challenges, leadership skills etc.

UNIT -1

Application of Grammar, Verbs Tense. Do as directed (active/passive, Direct/ Indirect, affirmative/ Negative/ Assertive/ Interrogative, Question tag, remove too, use of article, preposition, conjunction, punctuation). Correct the errors from the sentences. Vocabulary Building (Synonyms/ Antonyms/ Homophones/ Use of contextual word in given paragraph)

UNIT-2

Introduction to communication Definition, Communication cycle Concept of Communication Process Formal Communication Formal: Types- a) Vertical Communication b) Horizontal Communication Informal: Types- Diagonal Communication Verbal Vs Non-Verbal



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Communication.

Verbal: Types a) Oral Communication b) Written Communication Non-Verbal: Types a) Body Language b) Graphic Language

UNIT-3

Principal of Effective Communication. Principal of Effective Communication, Communication barriers and how to overcome them.

UNIT-4

Developing Effective message: Thinking about purpose, knowing the audience, structuring the message, selecting proper channels, minimize barriers and facilitating feedback. (Example: Writing articles for newspapers, magazines)

Suggested books:

1. Contemporary English Grammar Structures and Composition; David Green, Macmillan
2. English Grammar and composition; R. C. Jain, Macmillan
3. Effective Technical Communication; M. Ashraf Rizvi, Tata McGraw Hill Companies
4. Developing communication Skills; Krushna Mohan, Meera Baneji, Macmillan.

Course outcomes:

At the end of this course, the participants will:

1. Develop basic speaking and writing skills including proper usage of language and vocabulary so that they can become highly confident and skilled speakers and writers.
2. Be informed of the latest trends in basic verbal activities such as presentations, facing interviews and other forms of oral communication.
3. Also develop skills of group presentation and communication in team.
4. Develop non-verbal communication such as proper use of body language and gestures.



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Diploma Engineering Semester-I

(Common to all Branches)

Course Code	DE106
Course Title	Engineering Graphics
Credits	2(L: 2 T: 0 P: 0)

Course Objectives:

- To understand the language of graphics which is used to express ideas, convey instructions while carrying out engineering jobs.
- To develop drafting and sketching skills, to know the applications of drawing equipment, and get familiarize with Indian Standards related to engineering drawings.
- To understand the basic commands and develop basic skills related to computer aided drafting, of how to draw, modify, and edit basic shapes (2D), using AUTOCAD.

UNIT – I

Basic elements of Drawing: Drawing Instruments and supporting materials: method to use them with applications. Convention of lines and their applications. Representative Fractions – reduced, enlarged and fullsize scales; Engineering Scales such as plain and diagonal scale. Dimensioning techniques as per SP-46:2003 – types and applications of chain, parallel and coordinate dimensioning. Geometrical and Tangency constructions. (Redraw the figure)

UNIT – II

Orthographic projections: Introduction of projections-orthographic, perspective, isometric and oblique: concept and applications. (No question to be asked in examination). Introduction to orthographic projection, First angle and Third angle method, their symbols. Conversion of pictorial view into Orthographic Views – object containing plain surfaces, slanting surfaces, slots, ribs, cylindrical surfaces. (use First Angle Projection method only)

UNIT – III

Isometric Projections: Introduction to isometric projections. Isometric scale and Natural scale.



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Isometric view and isometric projection. Illustrative problems related to objects containing lines, circles and arcs shape only. Conversion of orthographic views into isometric view/projection.

UNIT – IV

Hand Sketches of engineering elements: Free hand sketches of machine elements: Thread profiles, nuts, bolts, studs, set screws, washer, Locking arrangements. (For branches other than mechanical Engineering, the teacher should select branch specific elements for free hand sketching) Free hand sketches of orthographic view (on squared graph paper) and isometric view (on isometric grid paper)

UNIT – V

Computer aided drafting interface: Computer Aided Drafting: concept. Hardware and various CAD software available. System requirements and understanding the interface. Components of AutoCAD software window: Title bar, standard tool bar, menu bar, object properties tool bar, draw tool bar, modify tool bar, cursor cross hair. Command window, status bar, drawing area, UCS icon. File features: new file, Saving the file, opening an existing drawing file, Creating templates, Quit. Setting up new drawing: Units, Limits, Grid, Snap. Undoing and redoing action.

Unit – VI

Computer aided drafting: Draw basic entities like Line, Circle, Arc, Polygon, Ellipse, Rectangle, Multiline, Poly Line. Method of Specifying points: Absolute coordinates, Relative Cartesian and Polar coordinates. Modify and edit commands like trim, extend, delete, copy, offset, array, block, layers. Dimensioning: Linear, Horizontal Vertical, Aligned, Rotated, Baseline, Continuous, Diameter, Radius, Angular Dimensions. Dim scale variable. Editing dimensions. Text: Single line Text, Multiline text. Standard sizes of sheet. Selecting Various plotting parameters such as Paper size, paper units, Drawing orientation, plot scale, plot offset, plot area, printpreview.



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Reference Books:

1. Bhatt, N. D. *Engineering Drawing*, Charotar Publishing House, Anand, Gujrat 2010; ISBN: 978-93- 80358-17-8
2. Dhawan, R. K. *Engineering Drawing*. S. Chand and Company, New Delhi; ISBN: 81-219-1431- 0.
3. Kulkarni, D. M.; Rastogi, A. P.; Sarkar, A. K. *Engineering Graphics with AutoCAD*. PHI Learning Private Limited-New Delhi (2010); ISBN: 978-8120337831.

Software/Learning Websites

1. <https://www.youtube.com/watch?v=TJ4jGyD-WCw>
2. https://www.youtube.com/watch?v=dmt6_n7Sgcg
3. <https://www.youtube.com/watch?v=MQScnLXL0M>
4. <https://www.youtube.com/watch?v=3WXPanCq9LI>

Course Outcomes

Following outcomes will be achieved:

- 1) Select and construct appropriate drawing scales, use drawing equipment's, and understand Indian Standards of engineering drawing
- 2) Draw views of given object and components, Sketch orthographic projections into isometric projections and vice versa.
- 3) Apply computer aided drafting tools to create 2D engineering drawings



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Diploma Engineering Semester-I (Common to all Branches)

Course Code	DE102
Course Title	Mathematics-I
Credits	3 (L: 3 T: 0 P: 0)

Course Objectives:

This course is designed to give a comprehensive coverage at an introductory level to the subject of Trigonometry, Differential Calculus and Basic elements of algebra.

UNIT-I

Trigonometry: Concept of angles, measurement of angles in degrees, grades and radians and their conversions, T-Ratios of Allied angles (without proof), Sum, difference formulae and their applications (without proof). Product formulae (Transformation of product to sum, difference and vice versa). T- Ratios of multiple angles, sub-multiple angles ($2A$, $3A$, $A/2$). Graphs of $\sin x$, $\cos x$, $\tan x$ and e^x

UNIT – II

Differential Calculus: Definition of function; Concept of limits. Four standard limits:

Differentiation by definition of $\sin x$ and $\cos x$, $\tan x$, $\log_a x$ Differentiation of sum, product and quotient of functions. Differentiation of function of a function. Differentiation of trigonometric a inverse trigonometric functions, Logarithmic differentiation, Exponential functions.

UNIT-III

Algebra: Complex Numbers: Definition, real and imaginary parts of a Complex number, polar and Cartesian, representation of a complex number and its conversion from one form



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to other, conjugate of a complex number, modulus and amplitude of a complex number
Addition, Subtraction, Multiplication and Division of a complex number. De-Moivre's
Theorem, its application.

UNIT – IV

Partial fractions: Definition of polynomial fraction proper & improper fractions and definition of partial fractions. To resolve proper fraction into partial fraction with denominator containing non- repeated linear factors, repeated linear factors and irreducible non-repeated quadratic factors. To resolve improper fraction into partial fraction

UNIT – V

Permutations and Combinations: Value of nP_r and nC_r

Binomial theorem: Binomial theorem (without proof) for positive integral index (expansion and general form); binomial theorem for any index (expansion without proof) first and second binomial approximation with applications to engineering problems

Reference Books:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi, 40th Edition, 2007.
2. G. B. Thomas, R. L. Finney, Calculus and Analytic Geometry, Addison Wesley, 9th Edition, 1995.
3. Reena Garg, Engineering Mathematics, Khanna Publishing House, New Delhi (Revised Ed. 2018)

Course Outcomes:

By the end of the course, the students are expected to learn

- (i) The students are expected to acquire necessary background in Trigonometry to appreciate the importance of the geometric study as well as for the calculation and the



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mathematical analysis.

- (ii) The ability to find the effects of changing conditions on a system.
- (iii) Complex numbers enter into studies of physical phenomena in ways that most people cannot imagine.
- (iv) The partial fraction decomposition lies in the fact that it provides an algorithm for computing the anti-derivative of a rational function.



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Diploma Engineering Semester-I

(Common to all Branches)

Course Code	DE157
Course Title	Workshop/Manufacturing Practices Lab
Credits	1.5(L: 0 T: 0 P: 3)

Course Objectives:

- To understand basic engineering processes for manufacturing and assembly.
- To understand, identify, select and use various marking, measuring, and holding, striking and cutting tools and equipment's
- To understand and interpret job drawings, produce jobs, and inspect the job for specified Dimensions
- To understand the various types of wiring systems and acquire skills in house wiring
- To understand, operate, control different machines and equipment's adopting safety practices

Details of Practical Content:

1. Carpentry:

- (i) Demonstration of different wood working tools / machines.
- (ii) Demonstration of different wood working processes, like planing, marking, chiseling, grooving, turning of wood etc.
- (iii) One simple job involving any one joint like mortise and tenon dovetail, bridle, half lap etc.

2. Fitting:

- (i) Demonstration of different fitting tools and drilling machines and power tools
- (ii) Demonstration of different operations like chipping, filing, drilling, tapping, sawing, cutting etc.
- (iii) One simple fitting job involving practice of chipping, filing, drilling, tapping, cutting etc.

3. Welding:

- (i) Demonstration of different welding tools / machines.
- (ii) Demonstration on Arc Welding, Gas Welding, MIG, MAG welding, gas cutting and rebuilding of broken parts with welding.
- (iii) One simple job involving butt and lap joint.

4. Sheet Metal Working:

- (i) Demonstration of different sheet metal tools / machines.



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- (ii) Demonstration of different sheet metal operations like sheet cutting, bending, edging, end curling, lancing, soldering, brazing, and riveting.
- (iii) One simple job involving sheet metal operations and soldering and riveting.

5. Electrical House Wiring:

- (i) Practice on simple lamp circuits
- (ii) One lamp controlled by one switch by surface conduit wiring,
- (iii) Lamp circuits- connection of lamp and socket by separate switches,
- (iv) Connection of Fluorescent lamp/tube light
- (v) simple lamp circuits-install bedroom lighting.
- (vi) Simple lamp circuits- install stair case wiring.

6. Demonstration:

- (i) Demonstration of measurement of Current, Voltage, Power and Energy.
- (ii) Demonstration of advance power tools, pneumatic tools, electrical wiring tools and accessories.
- (iii) Tools for Cutting and drilling

References:

1. S.K. Hajara Chaudhary, Workshop Technology, Media Promoters and Publishers, New Delhi, 2015
2. B.S. Raghuwanshi, Workshop Technology, Dhanpat Rai and sons, New Delhi 2014
3. K. Venkat Reddy, Workshop Practice Manual, BS Publications, Hyderabad

Learning Outcome

After undergoing this lab work, the student will be able to:

- Select right kind of measuring tools (Meter scale, Vernier caliper, Screw gauge, Spherometer etc.) for determining dimensions of physical quantities and make measurements with accuracy and precision.
- Differentiate various shapes and determine dimensions of plane, curved and regular surfaces/bodies.
- Apply and Verify laws of forces and determine resultant force acting on a body.
- Appreciate role of friction and measure co-efficient of friction between different surfaces.
- Describe and verify Hook's law and determine force constant of spring body.



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(Diploma EEE)

Diploma Engineering Semester-I

(Common to all Branches)

Course Code	DE105
Course Title	Fundamentals of Computer
Credits	2(L: 2 T: 0 P: 0)

Course Objective

1. Understand the basic concept of input and output device and memory.
2. Learn the functional units and classify types of computer
3. Learn about the different operating system.

Unit- 1

Basic Internet skills: Understanding browser, efficient use of search engines, awareness about Digital India portals (state and national portals) and college portals. General understanding of various computer hardware Components — CPU, Memory, Display, Keyboard, Mouse, HDD and other Peripheral Devices.

Unit- 2

OS Installation (Linux and MS Windows), Unix Shell and Commands, vi editor.

Unit- 3

HTML4, CSS, making basic personal webpage.

Unit- 4

Office Tools: Open Office Writer, Open Office Spreadsheet (Calc), Open Office Impress.

Unit-5

Information security best practices. Class lectures will only introduce the topic or demonstrate tool, actual learning will take place in the Lab by practicing

Course Outcome:

After this course the student will be able to:

1. Able to handle computer system for technical use
2. Differentiate between system and application
3. Able to work on MS-office
4. Able to install different software.



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Credit System and Marks Distribution (2023 batch):-

Semester-II												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	BSC	DE201	Mathematics – II	3	0	0	3	30	70	21	100	35
2	BSC	DE202	Applied Physics – II	3	0	0	3	30	70	21	100	35
3	ESC	DE203	Fundamentals of Electrical & Electronics Engineering	2	0	0	2	30	70	21	100	35
4	ESC	DE204	Engineering Mechanics	2	0	0	2	30	70	21	100	35
5	Humanities & Social Sc. Course	DE205	Communication Skills in EnglishII	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	BSC	DE252	Applied Physics –II Lab	0	0	2	1				50	25
2	ESC	DE253	Fundamentals of Electrical & Electronics Engineering Lab	0	0	2	1				50	25
3	ESC	DE254	Engineering Mechanics Lab	0	0	2	1				50	25
TOTAL							16					



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Diploma Engineering Semester-II

(Common to all Branches)

Course Code	DE201
Course Title	Mathematics-II
Credits	3(L: 3 T: 0 P: 0)

Course Objectives:

1. Describe memberships of sets, including the empty set, using proper notation, and decide whether given items are members and determine the cardinality of a given set.
2. Describe the relations between sets regarding membership, equality, subset, and proper subset, using proper notation.

UNIT-I

COORDINATE GEOMETRY:

Co-ordinate System: Cartesian and Polar. Distance, Division, Area of a triangle. □ Locus of a point and its equation. **Slope of St. Line** □ Angle between two St. lines, □ Parallel and perpendicular St. lines. □ **Standard and general equation of St. line.** □ Point of intersection of two st lines.

UNIT-II

INTEGRAL CALCULUS :Definition as a inverse process of differentiation, Standard Results (including inverse function), **Methods of Integration**; Substitution ,Integration by parts, Breaking up into partial fraction, Concept of Definite I

UNIT-III

NUMERICAL SOLUTION:-Bisection method, Regula-Falsi method and Newton Raphson method Numerical Solution of Simultaneous Equations □ Gauss elimination method Iterative methods-Gauss Seidal and Jacobi's method

UNIT-IV

STATISTICS AND PROBABILITY:-Measures of Central tendency (mean, median, mode) for ungrouped and grouped frequency distribution., Graphical representation to find mode and median, Measures of Dispersion such as range, mean deviation, Standard Deviation, Variance and coefficient of variation. □ Comparison of two sets of observations. □ Definition of random



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experiment, sample space, event, Occurrence of event and types of events (impossible, mutually exclusive, exhaustive, equally likely). □ Definition of Prob

UNIT-V

APPLICATION OF DERIVATIVES Geometrical meaning of Derivative, Equation of tangent and Normal, Rates and Motion, Maxima and minima, Radius of Curvature

References:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi, 40th Edition, 2007.
2. G. B. Thomas, R. L. Finney, Calculus and Analytic Geometry, Addison Wesley, 9th Edition, 1995.
3. Reena Garg, Engineering Mathematics, Khanna Publishing House, New Delhi (Revised Ed. 2018)
4. S.P. Deshpande Mathematics for Polytechnic Pune Vidyarthi Griha Prakashan, Pune.

Course Outcomes

1. Be able to draw and interpret Venn diagrams of set relations and operations and use Venn diagrams to solve problems.
2. Recognize when set theory is applicable to real-life situations, solve real-life problems, and communicate real-life problems and solutions to others.



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Diploma Engineering Semester-II

(Common to all Branches)

Course Code	DE202
Course Title	Applied Physics-II
Credits	3(L: 3 T: 0 P: 0)

Course Objectives:

Applied physics aims to give an understanding of this world both by observation and prediction of the way in which the object behaves; this course helps the student to solve board-based engineering problems.

Unit – 1

Wave motion and its applications:

Wave motion, transverse and longitudinal waves with examples, definitions of wave velocity, frequency and wave length and their relationship, Sound and light waves and their properties, Wave equation, Amplitude, Phase, Phase difference, principle of superposition of waves and beat formation. Simple Harmonic Motion (SHM): definition, expression for displacement, velocity, acceleration, time period, frequency etc. Simple harmonic progressive wave and energy transfer, study of vibration of cantilever and determination of its time period, Free, forced and resonant vibrations with examples.

Acoustics of buildings – reverberation, reverberation time, echo, noise, coefficient of absorption of sound, methods to control reverberation time and their applications, Ultrasonic waves – Introduction and properties, engineering and medical applications of ultrasonic.

Unit – 2

Optics: Basic optical laws; reflection and refraction, refractive index, Images and image formation by mirrors, lens and thin lenses, lens formula, power of lens, magnification and defects. Total internal reflection, Critical angle and conditions for total internal reflection, applications of total internal reflection in optical fiber. Optical Instruments; simple and compound microscope, astronomical telescope in normal adjustment, magnifying power, resolving power, uses of microscope and telescope, optical projection systems.



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Unit – 3

Electrostatics: Coulombs law, unit of charge, Electric field, Electric lines of force and their properties, Electric flux, Electric potential and potential difference, Gauss law: Application of Gauss law to find electric field intensity of straight charged conductor, plane charged sheet and charged sphere. Capacitor and its working, Types of capacitors, Capacitance and its units. Capacitance of a parallel plate capacitor, Series and parallel combination of capacitors (related numerical), dielectric and its effect on capacitance, dielectric break down.

Unit – 4

Current Electricity: Electric Current and its units, Direct and alternating current, Resistance and its units, Specific resistance, Conductance, Specific conductance, Series and parallel combination of resistances. Factors affecting resistance of a wire, carbon resistances and colour coding. Ohm's law and its verification, Kirchhoff's laws, Wheatstone bridge and its applications (slide wire bridge only), Concept of terminal potential difference and Electro motive force (EMF) Heating effect of current, Electric power, Electric energy and its units (related numerical problems), Advantages of Electric Energy over other forms of energy.

Unit – 5

Electromagnetism: Types of magnetic materials; dia, para and ferromagnetic with their properties, Magnetic field and its units, magnetic intensity, magnetic lines of force, magnetic flux and units, magnetization. Concept of electromagnetic induction, Faraday's Laws, Lorentz force (force on moving charge in magnetic field). Force on current carrying conductor, force on rectangular coil placed in magnetic field. Moving coil galvanometer; principle, construction and working, Conversion of a galvanometer into ammeter and voltmeter.

Semiconductor Physics: Energy bands in solids, Types of materials (insulator, semi- conductor, conductor), intrinsic and extrinsic semiconductors, p-n junction, junction diode and V-I characteristics, types of junction diodes. Diode as rectifier – half wave and full wave rectifier (centre taped). Transistor; description and three terminals, Types- pnp and npn, some electronic applications (list only). Photocells, Solar cells; working principle and engineering applications.



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Text Books:

Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi

Reference Books:

Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi

Course Outcomes:

After undergoing through syllabus, the students will be able to

- 1) Explain the wave motion, simple harmonic motion, polarization of wave etc.
- 2) Explain the ultrasonic waves and engineering, medical and industrial application of ultra sound.
- 3) Apply the knowledge of diodes, led, power adapters in electronic circuit.
- 4) Explain the current as flow of charge, concept power and resistance. Etc.
- 5) Express the nano science and technique and their effects on nanosecond and science and their impact on society. Etc.



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Diploma Engineering Semester-II

(Common to all Branches)

Course Code	DE252
Course Title	Applied Physics Lab-II
Credits	1 (L: 0 T: 0 P: 2)

Course Objectives:

Concrete use of physical principles and analysis in various fields of engineering and technology is very prominence. The course aims to supplement the factual knowledge gained in the lecture by first hand manipulation of apparatus. This will develop scientific temper and help to apply the basic concepts and principles in solving engineering and technology based problems.

List of Practical's/Activities: (To perform minimum 10 Practicals)

1. To determine and verify the time period of a cantilever.
2. To determine velocity of ultrasonic in different liquids using ultrasonic interferometer.
3. To verify laws of reflection from a plane mirror/ interface.
4. To verify laws of refraction (Snell's law) using a glass slab.
5. To determine focal length and magnifying power of a convex lens.
6. To verify Ohm's law by plotting graph between current and potential difference.
7. To verify laws of resistances in series and parallel combination.
8. To find the frequency of AC main using electrical vibrator.
9. To verify Kirchhoff's law using electric circuits.
10. To study the dependence of capacitance of a parallel plate capacitor on various factors and determines permittivity of air at a place.
11. To find resistance of a galvanometer by half deflection method.



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12. To convert a galvanometer into an ammeter.
13. To convert a galvanometer into a voltmeter.
14. To draw V-I characteristics of a semiconductor diode (Ge, Si) and determine its knee volt-age.
15. To verify inverse square law of radiations using a photo-electric cell.

Recommended Books:

1. Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi
2. Comprehensive Practical Physics, Vol, I & II, JN Jaiswal, Laxmi Publications (P) Ltd., New Delhi
3. Practical Physics by C. L. Arora, S. Chand & Company Ltd.

Course Outcome:

After undergoing this subject, the student will be able to;

1. Apply concept of vibrations and determine the time period of vibrating objects.
2. Verify optical laws; reflection, refraction from plane interfaces and surfaces.
3. Apply knowledge of optics to determine focal length and magnifying power of optical lenses.
4. Understand uses of electrical components and meters and verify Ohm's law for flow of current.



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(Diploma EEE)

Diploma Engineering Semester-II

(Common to all Branches)

Course Code	DE203
Course Title	Fundamental of Electrical & Electronics Engineering
Credits	1(L: 0T: 0 P: 2)

Course Objectives:

1. To familiarize students with basic electrical and electronic components and their functions
2. To make the students aware with AC and DC circuits and their analysis.

Unit-I

Overview of Electronic Components & Signals: Passive Active Components: Resistances, Capacitors, Inductors, Signals: DC/AC, voltage/current, periodic/non-periodic signals, average, rms, peak values, different types of signal waveforms, Ideal/non-ideal voltage/current sources, independent/dependent voltage current sources.

Unit- II

Diodes, Transistors, FET, MOS and CMOS and their Applications, Overview of Digital Electronics: Introduction to Boolean Algebra, Electronic Implementation of Boolean Operations, OP AMP 741

Unit- III

Electric and Magnetic Circuits: EMF, Current, Potential Difference, Power and Energy; M.M.F, magnetic force, permeability, hysteresis loop, reluctance, leakage factor and BH curve; Electromagnetic induction, Faraday's laws of electromagnetic induction, Lenz's law

Unit-IV

A.C. Circuits: Cycle, Frequency, Periodic time, Amplitude, Angular velocity, RMS value, Average value, Form Factor Peak Factor, impedance, phase angle, and power factor; Mathematical and phasor representation of alternating emf and current; Voltage and Current relationship in Star and Delta connections; A.C in resistors, inductors and capacitors; A.C in R-L series, R-C series, R-L-C



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series and parallel circuits; Power in A. C. Circuits, power triangle, Introduction Overview and applications of Electrical Machines DC Motor, DC Generator, AC Generator, Induction Motor and Transformer.

Text Books:

1. Basic Electrical Engineering I.J. Nagrath and D.P. Kothari, 2nd Edition, TMH, Delhi.
2. Basic Electrical Engineering by UA Bakshi and AV Bakshi Technical Publications Pune.

Reference Books:

1. Ritu Sahdev, Basic Electrical Engineering, Khanna Publishing House
2. Mittle and Mittal, Basic Electrical Engineering, McGraw Education, New Delhi, 2015, ISBN:978-0-07-0088572-5
3. Saxena, S. B. Lal, Fundamentals of Electrical Engineering, Cambridge University Press, latest edition ISBN: 9781107464353

Course Outcomes:

After completing the course students will be able to

1. Discuss about basic electrical and electronic devices
2. Solve network problems of AC and DC circuits
3. Can design basic analog and digital circuits.



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(Diploma EEE)

Diploma Engineering Semester-II

(Common to all Branches)

Course Code	DE253
Course Title	Fundamental of Electrical & Electronics Engineering Lab
Credits	1(L: 0 T: 0 P: 2)

Course Objectives:

1. To make the students able to do design and make connection for basic electrical circuit.
2. To enable students to verify the characteristics of the electrical and electronic circuits practically
3. To have sound practice of electrical safety.

List of Experiments

1. To study the various components of Electrical and Electronics.
2. To determine the stated value of a resistor by interpreting the color code indicated on the resistor.
3. To measure resistance using ohmmeter
4. To study and verify the series and parallel Circuit.
5. To study and plot V-I Characteristics of Silicon P-N Junction Diode.
6. To study the dependence of current on the potential difference across a resistor and determine its resistance. Also plot a graph between V and I.
7. To study the Inverting Amplifier and Non-Inverting Amplifier op-amp circuits.
8. To study the basic logic gates: AND, OR, NOT
9. To Study the Characteristics of Half – Wave Rectifier.
10. To study the Characteristics of Light Emitting Diode

Course Outcomes:

1. Students will be able to analyze and design electrical and electronic circuits practically.
2. Students will be able to study the characteristics of basic electrical circuits.
3. Students will be able to follow electrical safety in all prospects.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Engineering Semester-II

(Common to all Branches)

Course Code	DE204
Course Title	Engineering Mechanics
Credits	2(L: 2 T: 0 P: 0)

Course Objectives:

Following are the objectives of this course:

1. To obtain resultant of various forces
2. To calculate support reactions through conditions of equilibrium for various structures
3. To understand role of friction in equilibrium problems
4. To know fundamental laws of machines and their applications to various engineering problems

Unit -I

Basics of mechanics and force system: Significance and relevance of Mechanics, Applied mechanics, Statics, Dynamics. Space, time, mass, particle, flexible body and rigid body. Scalar and vector quantity, Units of measurement (SI units) - Fundamental units and derived units. Force – unit, representation as a vector and by Bow's notation, characteristics and effects of a force, Principle of transmissibility of force, Force system and its classification. Resolution of a force - Orthogonal components of a force, moment of a force, Varignon's Theorem. Composition of forces – Resultant, analytical method for determination of resultant for concurrent, non-concurrent and parallel co-planar force systems – Law of triangle, parallelogram and polygon of forces.

Unit- II

Equilibrium: Equilibrium and Equilibrant, Free body and Free body diagram, Analytical and graphical methods of analyzing equilibrium Lami's Theorem – statement and explanation, Application for various engineering problems. Types of beams, supports (simple, hinged, roller and fixed) and loads acting on beam (vertical and inclined point load, uniformly distributed load, couple), Beam reaction for cantilever, simply supported beam with or without overhang – subjected to combination of Point load and uniformly distributed load. Beam reaction graphically for simply supported beam



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subjected to vertical point loads only.

Unit– III

Friction: Friction and its relevance in engineering, types and laws of friction, limiting equilibrium, limiting

friction, co-efficient of friction, angle of friction, angle of repose, relation between co-efficient of friction and angle of friction. Equilibrium of bodies on level surface subjected to force parallel and inclined to plane.

Equilibrium of bodies on inclined plane subjected to force parallel to the plane only.

Unit– IV

Centroid and center of gravity: Centroid of geometrical plane figures (square, rectangle, triangle, circle, semi-circle, quarter circle) Centroid of composite figures composed of not more than three geometrical figures Centre of Gravity of simple solids (Cube, cuboid, cone, cylinder, sphere, hemisphere) Centre of Gravity of composite solids composed of not more than two simple solids.

Unit – V

Simple lifting machine: Simple lifting machine, load, effort, mechanical advantage, applications and advantages. Velocity ratio, efficiency of machines, law of machine. Ideal machine, friction in machine, maximum Mechanical advantage and efficiency, reversible and non-reversible machines, conditions for reversibility Velocity ratios of Simple axle and wheel, Differential axle and wheel, Worm and worm wheel,

Single purchase and double purchase crab winch, Simple screw jack, Weston's differential pulley block, geared pulley block.

Reference Books:

1. D.S. Bedi, Engineering Mechanics, Khanna Publications, New Delhi (2008)
2. Khurmi, R.S., Applied Mechanics, S. Chand & Co. New Delhi.
3. Bansal R K, A text book of Engineering Mechanics, Laxmi Publications.
4. Ramamrutham, Engineering Mechanics, S. Chand & Co. New Delhi.



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Course outcomes:

After completing this course, student will be able to:

1. Identify the force systems for given conditions by applying the basics of mechanics.
2. Determine unknown force(s) of different engineering systems.
3. Apply the principles of friction in various conditions for useful purposes.
4. Find the centroid and center of gravity of various components in engineering systems.
5. Select the relevant simple lifting machine(s) for given purposes.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING (Diploma EEE)

Diploma Engineering Semester-II (Common to all Branches)

Course Code	DE254
Course Title	Engineering Mechanics lab
Credits	1(L: 0 T: 0 P: 2)

Course Objectives:

Following are the objectives of this course:

- 1) To obtain resultant of various forces
- 2) To calculate support reactions through conditions of equilibrium for various structures
- 3) To understand role of friction in equilibrium problems
- 4) To know fundamental laws of machines and their applications to various engineering problems

List of Practical to be performed:

1. To study various equipments related to Engineering Mechanics.
2. To find the M.A., V.R., Efficiency and law of machine for Differential Axle and Wheel.
3. To find the M.A., V.R., Efficiency and law of machine for Simple Screw Jack.
4. Derive Law of machine using Worm and worm wheel.
5. Derive Law of machine using Single purchase crab.
6. Derive Law of machine using double purchase crab.
7. Derive Law of machine using Weston's differential or wormed geared pulley block.
8. Determine resultant of concurrent force system applying Law of Polygon of forces using forcetable.
9. Determine resultant of concurrent force system graphically.
10. Determine resultant of parallel force system graphically.



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Reference Books:

1. Bedi D.S., Engineering Mechanics, Khanna Publishing House
2. Khurmi, R.S., Applied Mechanics, S.Chand & Co. New Delhi.
3. Bansal R K, A text book of Engineering Mechanics, Laxmi Publications.
4. Ramamrutham, Engineering Mechanics, S.,S Chand & Co. New Delhi.

Course outcomes:

After completing this course, student will be able to

1. Identify the force systems for given conditions by applying the basics of mechanics.
2. Determine unknown force(s) of different engineering systems.
3. Apply the principles of friction in various conditions for useful purposes.
4. Find the centroid and center of gravity of various components in engineering systems.
5. Select the relevant simple lifting machine(s) for given purposes.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Engineering Semester-II

(Common to all Branches)

Course Code	DE205
Course Title	Communication Skills-2
Credits	3(L: 3T: 0 P: 0)

Course objectives:

1. To make students able to learn speak read and write the Basic English
2. To develop effective communication
3. To write business letters applications etc.

UNIT -1

Formal written skills Office Drafting: Circular, Notice and Memo Job Application with resume. Business correspondence: Enquiry, Order letter, Complaint letter and Adjustment letter.

Report writing: Accident report, fall in production, Progress report, Investigation report.

UNIT -2

Principles of Effective Communication Principles of Effective listening/Speaking Communication
Barriers, Overcoming Barrier

UNIT -3

Soft skill development

Speaking skill

Introduction to Group discussion Process of Group Discussion Leadership skill

Instant public speaking

UNIT -4

Etiquettes and Body Language Telephone etiquettes listening/Speaking Problems of telephonic conversation Verbal /oral etiquettes

Physical appearance



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(Diploma EEE)

Eye contact/ Body Language Group Discussion

Suggested Books:

1. Contemporary English Grammar Structures and Composition; David Green, Macmillan
2. English Grammar and composition; R. C. Jain, Macmillan
3. Effective Technical Communication; M. Ashraf Rizvi, Tata McGraw Hill Companies
4. Developing Communication Skills; Krushna Mohan, Meera Baneji, Macmillan

Course outcomes:

Students will be able to

1. Communicate effectively publically
2. Reading writing basic English will be improved so that it will be helpful in professional work.



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Credit System and Marks Distribution (2023 batch): -

Semester-III												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	BSC	DEEP301	Mathematics – III	3	0	0	3	30	70	21	100	35
2	ESC	DEEP302	Object Oriented Programming with C++	2	0	0	2	30	70	21	100	35
3	PCC	DEEP303	Network Theory	3	0	0	3	30	70	21	100	35
4	PCC	DEEP304	Electrical and Electronic Measurement and Instrumentation	3	0	0	3	30	70	21	100	35
5	PCC	DEEP305	Principal of Digital Electronics	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	PCC	DEEP351	Network Theory Lab	0	0	2	1				50	25
2	PCC	DEEP352	Electrical and Electronic Measurement and Instrumentation Lab	0	0	2	1				50	25
3	PCC	DEEP353	Principal of Digital Electronics Lab	0	0	2	1				50	25
4	ESC	DEEP354	Object Oriented Programming with C++ Lab	0	0	2	1				50	25
6	IN		Summer Internship (4 weeks) after 2 nd sem	0	0	4	2				50	25
TOTAL							20					



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma in Electrical and Electronics Engineering

Semester – III

(Common to all Branches)

Course Code	DEEP301
Course Title	Applied Mathematics-III
Credits	3 (L: 3 T: 0 P: 0)

Course Objectives:

This course is designed to give a comprehensive coverage at an introductory level to the subject of Trigonometry, Differential Calculus and Basic elements of algebra. basic transformation and analysis in time domain and frequency domain.

Unit -I.

Integration:

Rules of integration (Integrals of sum, difference, scalar multiplication). Methods of Integration.

Integration by substitution Integration of rational functions. Integration by partial fractions.

Integration by trigonometric transformation. Integration by parts. Definite Integration. Definition of definite integral. Properties of definite integral with simple problems. Application of integration

Area under the curve. Area between two curves. Mean and RMS values.

Unit – II.

Differential Equation

Definition of differential equation, order and degree of differential equation. Formation of differential equation for function containing single constant. Solution of differential equations of first order and first degree such as variable separable type, reducible to Variable separable, Homogeneous, Non-homogeneous, Exact, Linear and Bernoulli equations. Applications of Differential equations.



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Unit III.

Laplace Transform

Definition of Laplace transform, Laplace transform of standard functions. Properties of Laplace transform such as Linearity, first shifting, second shifting, multiplication by tn , division by t .

Unit- IV.

Fourier Series

Definition of Fourier series (Euler's formula). Series expansion of continuous functions in the intervals Series expansions of even and odd functions. Half range series.

Unit V.

Numerical Methods

Solution of algebraic equation Bisection method. Regularfalsi method.,Newton - Raphson method.Solution of simultaneous equations containing 2 and 3 unknowns Gauss elimination method.

Reference books:

1. Mathematics for polytechnic, S. P. Deshpande , Pune Vidyarthi Griha Prakashan, Pune.
2. Calculus: single variable, Robert T. Smith, Tata McGraw Hill.
3. Higher Engineering Mathematics, B. S Grewal, Khanna Publication, New Delhi

Course Outcomes:

By the end of the course the students are expected to learn

1. the students are expected to acquire necessary background in Determinants and Matrices so as to appreciate the importance of the Determinants are the factors that scale different parameterizations so that they all produce same overall integrals, i.e. they are capable of encoding the inherent geometry of the original shape.
2. the cumulative effect of the original quantity or equation is the Integration
3. the coordinate geometry provides a connection between algebra and geometry through graphs of lines and curv



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma in Electrical and Electronics Engineering

Semester – III

Course Code	DEEP302
Course Title	Object Oriented Programming with C++
Credits	2(L: 2 T: 0 P: 0)

Course Objectives:

1. Understand the features of C++ supporting object-oriented programming
2. Understand the relative merits of C++ as an object-oriented programming language
3. Understand how to produce object-oriented software using C++
4. Understand how to apply the major object-oriented concepts to implement object-oriented programs in C++, encapsulation, inheritance and polymorphism
5. Understand advanced features of C++ specifically stream I/O, templates and operator overloading

Unit- I

Object oriented Programming: Basic concept of oops, application of OOPs Beginning with C++: What is C++, structure of C++ program, creating, compiling, linking & executing a C++ program, Tokens, expressions & control structures, keywords, identifiers, basic data types, user-defined data types, derived data types, symbolic constants, type compatibility, variable declaration, dynamic initialization of variables, reference variables. Constant and literals, operators in C++, control statements, decision making statements

UNIT - II

Functions in C++: Defining a Function, Function Declarations, The Main Function, Function Prototyping, Call by Reference, Return by Reference, Inline Function, default argument, friend functions



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(Diploma EEE)

UNIT – III

Classes and objects: Introduction, specifying a Class, defining member Functions, C++ Program with Class, Nesting of Member functions, Private member functions, Memory Allocation for Objects, Static Data members, Static Member Functions, Arrays within a Class, Arrays of Objects, Objects as Function Arguments, Friendly Functions, Returning Objects.

UNIT - IV

Pointers: Declaration and initializing, Manipulation of pointers, pointers Expression and Pointer Arithmetic, Pointer with Arrays, Arrays of Pointers, Pointers to objects, this pointers, Arrays of Pointers to Objects, Data Abstraction, data Encapsulation in c++

UNIT - V

Constructors and Destructors Constructors, Parameterized Constructors, Multiple Constructors in a class, Copy constructor, Destructors, Inheritance: defining derived classes, single inheritance, multilevel inheritance, multiple inheritance, hierarchical inheritance, hybrid inheritance, virtual base class, abstract classes, constructors in derived classes, member class, nesting of class. Exception handling: Principle of Exception handling, Exception handling mechanism, multiple catch, Nested try, Rethrowing the Exception.

Text Books:

1. Yashvant kanetkar ,Let us C++,BPB Publications,2019
2. Sumit Arora, Computer Science with C++,Dhampat Rai,9th edition,2017
3. Bjarne Stroustrup ,The C++ Programming Language,Addison-wesley, 4th edition,2013
4. E. Balagurusamy ,Object-Oriented Programming with C++, 7th Edition, 2017

Reference Books:

1. Robert Lafore, “Object Oriented Programming in C++”, Galgotia Publication Pvt. Ltd,4 th edition, New Delhi, 2002
2. Herbert Schildt , “ C++ : The Complete Reference”,TMH ,NewDelhi,2003.
3. Ashok N Kamathane, “Object Oriented Programming with ANSI & Turbo C++”, Pearson Education, New Delhi, 2003.
4. Stanley B Lippman and Josee Lajoie, “C++ Primer”, Pearson Education, ND, 2001.
5. Venugopal K R, Rajkumar Buyya and Ravishankar T,” Mastering C++”, TMH, ND, 2006.



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Web Link:

<https://spoken-tutorial.org/>

Suggested Readings:

1. Object Oriented Programming in C++ by E. Balagruswamy
2. Object Oriented Programming in C++ Saurav Sahay Oxford University Press.
3. Object Oriented Programming in C++ R Rajaram New Age International Publishers 2nd Edition.
4. Object Oriented Programming in C++ by Robert Lafore Techmedia Publication

Course Outcome

- Describe OOPs concepts.
- Use functions and pointers in your C++ program.
- Understand tokens, expressions, and control structures.
- Explain arrays and strings and create programs using them.
- Describe and use constructors and destructors.
- Understand and employ file management.



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(Diploma EEE)

Diploma in Electrical and Electronics Engineering

Course Objectives:

Course Code	DEEP354
Course Title	Object Oriented Programming with c++Lab
Credits	1(L: T: 0 P: 2)

- Programming basics and the fundamentals of C++
- Data types in C++ • Mathematical and logical operations
- Using if statement and loops
- Arranging data in arrays
- Implementing pointers
- File management and dynamic memory allocation

SYLLABUS

List of Programs as Assignments:

1. Write an interactive program that will read in a +ve integer value and determine the following
i) If the integer is a prime number ii) If the integer is a Fibonacci number
2. WAP to compute $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$. Continue adding successive terms in the series until the value of the next term becomes smaller (in magnitude) than 10^{-5} . Test the program for $x = 1$, $x = 2$, and $x = 3$. In each case display the number of terms used to obtain the final answer.
3. WAP to generate every 3rd integer beginning with $I = 2$ and continue for all integers that are less than 150. Calculate the sum of those integers that are evenly divisible by 5.
4. WAP to find whether a given year is a leap year or not. Modify it to generate a list of leap years between two year limits given by user.
5. WAP to display the following pattern: 11 11 10 11 11 10 9 10 11 11 10 9 8 9 10 11



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6. Using Ternary / Conditional operator find the greatest among 3 numbers.
7. WAP to convert a decimal number into an equivalent number of the input base. Test your program for base 2,8,10 & 16.
8. WAP to read a number n, and print it out digit-by-digit, as a series of words. For e.g. 123 would be printed as “one two three”.
9. WAP to check whether any input +ve integer is palindrome or not.
10. WAP to simulate a simple calculator (+ - / * %) that takes two operands and an operator as input and displays the result.
11. WAP to find the GCD of two input +ve integer numbers.
12. WAP to swap the values of two variables without using a third variable.
13. Read a line of mixed text, and then write it out with all lower case and uppercase letters reversed, all digits replaced by 0s and all other characters (non-letters and non- digits) replaced by „*“.
14. WAP to find the product of two matrices A and B. Display the source matrices and product matrix C in matrix format.
15. WAP to find whether a given matrix is a triangular matrix or not.
16. WAP to find the transpose of a matrix. Display the source and the transposed matrix in matrix format.
17. Implement Prob. No. – 14 to 16 using functions for reading, manipulating and displaying the corresponding matrices in matrix form.
18. WAP to sort a list of strings alphabetically using a 2-dim. Character array
19. WAP to display the row sum and the column – sum of an input 2- dim. Matrix. Display the source matrix with row and column sum.
20. Write a recursive function to calculate $S = 2 + 4 + 6 + 8 + \dots + 2N$. Implement the function in a complete C program.
21. Write a function that accepts two arguments an array and its size n. It performs Bubble up sort on the array elements. Using indirection operator „*“ implement this in a complete C program.



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Display the source and the sorted array.

22. Using pointer, write a function that receives a character string and a character as argument. Delete all occurrences of this character in the string. The function should return corrected string with no holes.

23. Write a function for reading character string using pointer. Calculate the length of the string (without using `strlen()`). Finally print the string in reverse order, using pointer.

24. Implement prob. No. 14 using pointers representation of 2 – dim. array.

25. Implement prob. No. 15 using pointer representation of 2 dim. array.

26. Implement prob. No. 16 using pointer representation of 2 dim. array.

27. WAP to sort a list of strings into alphabetical order using array of pointers.

28. Create records of 60 students, where each record has fields-name, roll No.

Course Outcome

- Describe OOPs concepts.
- Use functions and pointers in your C++ program.
- Understand tokens, expressions, and control structures.
- Explain arrays and strings and create programs using them.
- Describe and use constructors and destructors.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – III

Course Code	DEEP303
Course Title	Network Theory
Credits	3(L: 3 T: 0 P: 0)

Course objective: to make students able to:

1. Understand basics of AC and DC circuits
2. Understand circuit theorems
3. Understanding transient responses.

Unit I

Principles of circuit Analysis (ONLY DC circuits): Mesh analysis.(Numerical). Node analysis with voltage current source (Numerical) Star/delta & Delta/star transformations. (Simple Numerical).Network Theorems (Statement, procedure, applications and areas of applications, Simple Numerical on DC Circuits): Superposition Theorem (for also AC but no numerical on AC). Thevenin's Theorem. Norton's Theorem. Source conversion/ideal voltage and current source. Maximum power transfer Theorem

UNIT II

Review of Basic concepts of electrical Circuit: Electric Circuit Elements R, L, C. Energy Sources. A.C. waveform and definition of various terms associated with it. Response of pure R, L, and C to AC supplies. Vector Representation of alternating quantity.

UNIT III

Single phase AC Circuits: Series AC circuits R-L, R-C and R-L-C circuits. Impedance, reactance, phasor diagram, impedance triangle, power factor, Average power, Apparent power, Reactive power, Power triangle (Numerical).

UNIT IV

Series Resonance, quality factor (Numerical) Parallel AC circuits R-L, R-C and R-L-C circuits. admittance, Susceptance, Solution by admittance method, phasor diagram and complex Algebra method. (Numerical)Parallel resonance, quality factor. Comparison of series and Parallel circuits.



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(Diploma EEE)

Initial Conditions in RC, RL circuits, Transient Analysis of RC, RL Circuits

UNIT V

Poly phase AC Circuits: Generation of three phases e.m.f. Phase sequence, polarity marking. Types of three-phase connections. Concept of unbalanced load and balanced load. Line, phase quantities and power in three phase system with balanced star and Delta connected load & their interrelationship. Advantages of poly phase circuits over single phase circuits.

TEXT BOOKS

1. Network & system - D. Roy Choudhury
2. Network & system - G.K. Mittal
3. Network & system - Vulkenberg
4. Network & system - Dacsur & Kuo
5. Network Theory - R.N. Pathak

Course outcomes: Students will be able to

1. Analyze the AC and DC circuits
2. Solve circuit numerical problems
3. Analyze two port network problem.
4. Analyze transients of circuits



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – III

Course Code	DEEP351
Course Title	Network Theory Lab
Credits	1(L: 0 T: 0 P: 2)

Course Objectives:

To make students able to:

1. Verify circuits theorems practically on breadboard
2. Verify two port networks practically.
3. Analyze AC Circuits and transient in circuits practically

List of Experiments (Expandable)

1. To verify Thevenin Theorem practically
2. To Verify Norton Theorem Practically.
3. To verify Superposition Theorem Practically,
4. To verify Maximum Power Transfer Theorem
5. To study the characteristics of RLC Series Resonance.
6. To study RC RL and RLC Transient circuits.

Course outcomes: Students will be able to

1. Design circuits on breadboard
2. Analyze and measure the circuit parameters.
3. Analyze transients in circuits.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – III

Course Code	DEEP304
Course Title	Electrical and Electronic Measurement and Instrumentation
Credits	3(L: 3 T: 0 P: 0)

Course objectives: to make students able to

1. Understand the static and dynamic characteristics of instruments
2. Analyze the principle of working of various electrical instruments
3. Analyze the working of transducers

Unit-1

Concepts of Measurement systems: Fundamental and derived units, Methods of Measurement, Classification of Instruments: Absolute and Secondary Instruments, Analog and Digital Instruments, Mechanical, Electrical and Electronics Instruments, Manual and Automatic Instruments. Measurement of Errors.

Introduction, Classification of Analog Instruments, Principle of Operation, Operating Torques, Constructional Details: Moving System, Controlling System, Damping System; Permanent Magnet Moving Coil Instrument, Extension of Range of PMMC Instruments, Moving Iron Instruments: Attraction-type, Repulsion-type, Torque equation of Moving-Iron Instruments.; Electrodynamometer-type instruments, Electrostatic Instruments, Induction-type instruments, Electro-thermal Instruments.

Unit -II

Instrument Transformers: Introduction, Advantages of Instrument Transformers, Current Transformers (CT), Theory of Current Transformers, Errors Introduced by CT.

Introduction, Power Measurement in dc and ac Circuits, Electrodynamometer type Wattmeter, Induction-type Wattmeter, Power Measurement in Poly- Phase Systems.

Unit-III

Measurement of resistance and AC Bridges: Introduction, Measurement of Medium Resistances: Ohmmeter Method, Voltmeter-Ammeter Method, Substitution Method, Wheatstone Bridge for Measuring Resistance, Measurement of Low Resistances, High Resistances Potentiometers Introduction: A Basic dc Potentiometer, Application of dc potentiometer: Measurement of current, High voltage, resistance, Power by potentiometer.; AC potentiometers. Advantage and Disadvantage of AC Potentiometers. Applications of ac Potentiometer.

AC Bridges: Introduction, Sources and Detectors in ac Bridges, Measurement of Self-Inductance: Maxwell's Inductance Bridge, Maxwell's Inductance- Capacitance Bridge, Anderson's Bridge.; M



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Unit-IV

CRO and Digital Meters: Introduction, Block Diagram of a Cathode Ray Tube, Electrostatic Deflection, Time Base Generator, Measurement of Electrical Quantities with CRO, Measurement of voltage, current, frequency, Phase Difference, Sampling Oscilloscope.

Introduction, Merits of Digital System over Analog, Performance Characteristics of Digital Meters, Digital Multimeter, Digital Frequency Meter, Digital Voltmeters. Applications of DVMs.

Books Recommended

1. Electrical & Electronics Measuring Instrument- Dhanpat Rai & Sons. A.K Sawhney
2. Electrical Measurement & Measuring Instrument-Khanna Publisher—Rejendra Prasad
3. Electrical Measurement & Measuring Instrument E.W.Golding
4. Electrical & Electronic Measurement by J B Gupta
5. Electrical & Electronic Measurement by Ryder
6. Electronic Instrumentation and Measurement by W D Cooper

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

- a) Check the working of the electrical measuring instrument.
- b) Use different types of measuring instruments for measuring voltage and current.
- c) Use different types of measuring instruments for measuring electric power
- d) Use different types of measuring instruments for measuring electric energy.
- e) Use different types of electrical instruments for measuring various ranges of electrical parameters.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – III

Course Code	DEEP352
Course Title	Electrical and Electronic Measurement and Instrumentation Lab
Credits	1(L: 0 T: 0 P: 2)

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

Use relevant measuring instrument in different electrical applications.

List of Experiments (Any 10)

1. Measurement of Current and Voltages by Low range ammeter and voltmeter respectively with shunt and multiplier.
2. Calibration of Wattmeter at various power factors by standard Wattmeter.
3. Measurement of active power in three phase balanced load by single wattmeter method.
4. Measurement of active and reactive power in three phase balanced load by two wattmeter method.
5. Measurement of single-phase power with 3 ammeters and 3 voltmeters.
6. Calibration of Energy meter at various power factors by standard energy meter.
7. Measurement of energy in single phase & three phase balanced load using Electronic Energy Meter.
8. Measurement of Low resistance by Kelvin's Double Bridge.
9. Measurement of Medium resistance by Wheatstone bridge.
10. Measurement of Insulation Resistance by Megger.
11. Measurement of Resistance, Voltage, Current, Voltage, Current in A.C & D. C. Circuit by using digital multimeter.
12. Measurement of A.C. Current by tongue tester.
13. Measurement of Circuit Parameters by LCR meter.
14. To measure linear displacement by LVDT and plot characteristics.



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15. Measurement of inductance by Maxwell Bridge.
16. Measurement of Capacitance by Schering Bridge.
17. Measurement of inductance by Hay's Bridge.

COURSE OUTCOMES:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

1. Check the working of the electrical measuring instrument.
2. Use different types of measuring instruments for measuring voltage and current.
3. Use different types of measuring instruments for measuring electric power



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – III

Course Code	DEEP305
Course Title	Principles of Digital Electronics
Credits	3 (L: 3 T: 0 P: 0)

Course objective: The objective of this course is to provide the fundamental concepts associated with the digital logic and circuit design. To introduce the basic concepts and laws involved in the Boolean algebra and logic families and digital circuits. To familiarize with the different number systems, logic gates, and combinational and sequential circuits utilized in the different digital circuits and systems. The course will help in design and analysis of the digital circuit and system.

Unit I

Introduction: Basic difference between analog and digital signals, applications and advantages of analog signals. Number System: Binary and hexadecimal number system, conversion from decimal and hexadecimal number system to binary and vice versa. BCD representation. Binary addition, subtraction, multiplication and division including binary points, BCD addition. One's and two's Compliment method for addition and Subtraction. Codes and Parity: Concept of codes, weighted and non-weighted codes, examples of 8421, BCD, excess-3 and gray codes. Concept of parity, single and double parity and error detection, Alpha numeric codes.

UNIT II

Logic Gates: Concept of negative and positive logic, definition, symbols and truth tables of NOT, AND, OR, NAND, EXOR gates. NAND and NOR Gates. Logic Simplification Postulates of Boolean Algebra, DeMorgan's theorem, various identities. Formulation of truth table and Boolean equation for simple problem. Implementation of Boolean (logic) equation with gates. Karnaugh's Map (up to 4 variables) and simple applications in developing combinational logic circuits. Logic Families: Logic family Classifications: Definition of SSI, MSI, LSI, VLSI. TTL and MOS families and their sub classification. Characteristics of TTL and MOS digital gates, delay, speed, noise margin, logic levels, power dissipation, fan in, fan out, power supply requirement and comparison between TTL and MOS families. Interfacing TTL and MOS IC's. Logic Circuits: Open collector, wired OR and totem pole output circuit operation for a TTL NAND gate. MOS circuit operation for standard gate.

UNIT III

Arithmetic Circuits: Half adder and full adder circuit, design and implementation. Half and full subtractor circuit, design and implementation. Basic function and block diagram of MUX and DEMUX. Latches and Flip flops: Concept and types of latches with their working and applications. Operation using waveforms and truth tables of RS, T, D, JK, Master/Slave JK flip flops. Counters: Binary counters: Divide by N ripple counters (including Design), Decade counter. Resettable and



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programmable counters, Down Counters, up/ down Counter. Synchronous Counter. Difference between Asynchronous and Synchronous counters: Ring Counter with timing diagram.

UNIT IV

Shift Register and Memories: Introduction and basic concepts including shift left and shift right.

Serial in parallel out, serial in serial out, parallel in serial out, parallel in parallel out. Universal shift register. Buffer register, Tristate Buffer register, Basic RAM cell, N X M bit RAM, static and dynamic RAM, basic idea of ROM, PROM, EPROM, AND EEPROM.

UNIT V

A/D and D/A Convertors: General principle of A/D and D/A conversion and brief idea of their applications. Binary resistor network and resistor ladder network methods of D/A conversion. Dual slope and successive approximation types of ADCs.

References:

1. A. Anand Kumar, Fundamentals of digital circuits, PHI
2. A K Maini, Digital Electronics, Wiley India
3. Thomas Blakeslee; Digital Design with standard MSI and LSI; Wiley Inter science.
4. Jain RP; Modern digital electronic

Course Outcomes:

At the end of this course, students will demonstrate the ability to

1. Understand basics of numbers system, Boolean algebra, K map etc.
2. Design Flip Flop, Shift registers, Counters
3. Design PLA , PAL, PLDS etc



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – III

Course Code	DEEP353
Course Title	Principle of Digital Electronics Lab
Credits	1(L: 0 T: 0 P: 2)

Course Objective

To know the concepts of Combinational circuits.

To understand the concepts of flip-flops, registers and counters

List of Experiments:

1. To verify the truth table of logic gates realize AND, OR, NOT gates.
2. To realize AND, OR gates using diodes and resistors.
3. To verify the Boolean algebra function using digital IC gates (consensus theorem).
4. To realize the function $F(A, B, C, D) = (C+D)(A+B)(B+D)$ neither using NOR gates.
5. Design a half/full adder circuit using FF for 2 bits.
6. Design a half/full sub tractor circuit using FF for 2 bits.
7. Design a binary to gray code converter.
8. Design BCD to seven segment display using 7447 IC.
9. Implement $F(A, B, C) = \Sigma(1, 3, 4, 5, 6)$ with a multiplexer.

Books Recommended:

1. Digital Principles—Tata McGrawHill (TMH)—Malvino & Leach
2. Modern Digital Electronics—TMH---R.P.Jain

Course Outcome

Learn the basics of gates.

Construct basic combinational circuits and verify their functionalities

Apply the design procedures to design basic sequential circuits

Learn about counters, Shift registers

To understand the basic digital circuits and to verify their operation



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Credit System and Marks Distribution:-

Semester-IV												
SL. No.	Category	SubjectCode	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	PCC	DEEP401	Power System-I	3	0	0	3	30	70	21	100	35
2	PCC	DEEP402	Electrical Machine-I	3	0	0	3	30	70	21	100	35
3	PCC	DEEP403	Applied Electronics	3	0	0	3	30	70	21	100	35
4	PCC	DEEP404	Electrical Estimation and Costing	3	0	0	3	30	70	21	100	35
5	PEC	DEEP(E)	Program Elective-I	3	0	0	3	30	70	21	100	35
6	PEC	DEEP(E)	Program Elective-II	3	0	0	3	30	70	21	100	35
Program Elective-I (Choose any one)												
1	PEC	DEEP405(E)	Signals and Systems	3	0	0	3	30	70	21	100	35
2	PEC	DEEP406(E)	Communication technology	3	0	0	3	30	70	21	100	35
3	PEC	DEEP407(E)	Illumination practices	3	0	0	3	30	70	21	100	35
Program Elective-II (Choose any one)												
1	PEC	DEEP408(E)	Utilization of Electrical Energy	3	0	0	3	30	70	21	100	35
2	PEC	DEEP409(E)	Electric Traction	3	0	0	3	30	70	21	100	35
3	PEC	DEEP410(E)	Biomass and Micro-Hydro power plant	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	PCC	DEEP451	Power System -I Lab	0	0	2	1				50	25
2	PCC	DEEP452	Electrical Machine I lab	0	0	2	1				50	25
3	PROJ		Minor Project	0	0	4	2				50	25
4			Essence of Indian Knowledge & Tradition	0	0	0	0				50	25
TOTAL							22					



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP401
Course Title	Power System-I
Credits	3(L: 3 T: 0 P: 0)

Course objective: to make students able to:

1. Understand the power system architecture
2. Understanding Line parameters and calculating it
3. Understand various effects like Corona effect, proximity effect etc
4. Understanding rating of electrical equipment.

Unit I

Introduction to transmission. Necessity of transmission of electricity. Classification & comparison of different transmission systems. Transmission Line Components. Introduction to line components. types of conductors-Copper, Aluminum & state their trade names. Solid, Stranded & bundled conductors. Line supports – requirements, types, and field of applications. Line insulators – requirements, types, and field of applications

Unit II

Distribution of potential over a string of suspension insulators. Concept of string efficiency, Methods of improving string efficiency. corona formation, advantages & disadvantages, factors affecting corona, important terms related to corona. Spacing between Conductors. Calculation of Span length & sag Calculation (Numerical based) Skin effect, proximity effect & Ferranti effect

Unit III

Introduction & Requirement. EHVAC Transmission, Reasons for adoption & limitations. HVDC Transmission – Advantages, Limitations. Introduction. Classification of indoor & outdoor substations. Advantages & Disadvantages. Selection & location of site. Main connection schemes. Equipment's circuit element of substations. In coming & outgoing lines, Transformers, CT&PT, Relays, CB's, fuses, Isolators, batteries, lightning arresters.



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Unit IV

Introduction & requirements. Classification of cables. Cable conductors. Cable construction. Cable insulation, Metallic sheathing & mechanical protection. Comparison with overhead lines Cable laying.

Reference books:

1. K.B. Raina, S.K.Bhattacharya , Electrical Design; Estimating and costing, New Age International (p) Limited, New Delhi
2. Surjit Singh, Electrical Estimating and costing, Dhanpat Rai and company, New Delhi
3. N. Alagappan, S. Ekambaram, Electrical Estimating and costing, Tata Mc Graw Hill Publication, New Delhi
4. S.L. Uappal, Electrical wiring Estimating and Costing, Khanna Publication.
5. B.D.Arora, Electrical wiring, Estimating, and costing R.B. Publication, New Delhi

Course outcomes: students will be able to

1. Analyze the structure of power system.
2. Understanding line parameters and losses.
3. Analyze rating of electrical accessories.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP402
Course Title	Electrical Machine-I
Credits	3(L: 3 T: 0 P: 0)

Course Objectives:

Electrical machines are in wide use in industries and other services. In this course DC generator, DC motors, single phase transformer, auto transformer and three phase transformer is covered in details.

1. To acquire knowledge of construction, and control of the DC machines.
2. To acquire knowledge of performance of DC machines and transformers of all types.

Unit 1. D.C. GENERATORS

Principles of operation Constructional feature, Armature winding, Back pitch, Front pitch, Resultant pitch and commutator pitch, Simple Lap and wave winding (problems on winding diagram), Different types of D.C. Machines, Shunt, Series and Compound machines, Armature reaction in D.C. machine and Commutation,

Methods of improving commutation (Resistance and emf commutation), Inter poles and compensating winding Characteristics of D.C. Generators and uses of Different types of D.C. Generators. Concept of critical resistance, causes of failure of Development of emf, Losses and efficiency of D.C. Machines, condition for maximum efficiency, Parallel operation of D.C. Generators.

Unit-2 D.C. MOTORS

D.C. Motor principles, Signification of back emf in D.C. Motor, Voltage equation of Motor, Torque (equation of armature torque and shaft torque), Performance characteristics of shunt, series and compound motors and their application.

Methods of starting shunt, series and compound Motors, study of starters (3-point, 4-point starters and Drum controller type, problems in starter, Speed control of D.C. shunt motors 2.7.1 Flux control



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Method 2.7.2 Armature voltage (rheostatic) control method.

Ward Leonard method Speed control of series motors — Flux control method and series parallel control method.

Efficiency of DC machine by brake test, Efficiency of DC machine by Swinburne's test, Losses & efficiency and condition for maximum power

Unit -3 SINGLE PHASE TRANSFORMER

Working principles, Transformer construction — Arrangement of core & winding in different types of trans- former - Brief ideas about transformer accessories such as conservator, tank, breather, and explosion vent etc., Care and Maintenance, EMF equation, Voltage transformation ratio, Transformer on no load and on load Phasor diagrams, Equivalent resistance, reactance and impedance, Phasor diagram of transformer with winding resistance and magnetic leakage

Equivalent circuit Approximate & exact voltage drop of a Transformer, Regulation at various loads and power factor,

Different types of losses in a Transformer, open circuit test , Short Circuit test, Efficiency, efficiency at different loads and power factors, condition for maximum efficiency, All day efficiency, Determination of load corresponding to maximum efficiency, Parallel operation of single phase transformer, Introduction to Autotransformer, its advantages and applications

Unit -4 INTRODUCTION TO THREE—PHASE TRANSFORMER

Construction, principle of operation, parallel operation, connections, applications. Various connections and groups, choice of connection star-delta connections, Scott connection, three phase to two phase conversion and vice versa, Applications. Parallel operation of three phase transformer its conditions.

Text Books:

1. Electrical Machines by SK Bhattacharya, Tata Mc Graw Hill, Education Pvt Ltd. New Delhi
2. Electrical Machine by B.L. Thareja, S. Chand Publication, New Delhi.



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Reference Books:

1. Electrical Machines by Nagrath and Kothari, Tata Mc Graw Hill, New Delhi
2. Electrical Machines by JB Gupta, SK Kataria and Sons, New Delhi
3. Electrical Machines by Smarajit Ghosh-Pearson Publishers, Delhi.
4. e-books/e-tools/relevant software to be used as recommended by AICTE/UBTE/NITTTR, Course Chandigarh

Outcomes:

Students will be able to

1. Analyze the working and construction and losses of single and three phase transformers
2. Analyze the working of DC Motors dc generators and its various aspects
3. Analyze the working of Induction motors and its various aspects



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(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP452
Course Title	Electrical Machine-I Lab
Credits	1(L: 0 T: 0 P: 2)

Course objectives:

1. To make students familiar with making connection of transformers-based circuits.
2. To make students perform experiments based on DC Motor, DC Generator and Induction motors
3. Making connection diagrams and following safety while doing experiments.

LIST OF EXPERIMENTS:

1. To study the construction of a D C Machine.
2. To study the construction of 3-point starter of a D C Machine
3. To Measure power in a 3 phases circuit by 3 wattmeter method.
4. To Measure power in a 3 phases circuit by two wattmeter method.
5. To study the construction of single-phase transformer.
6. To study the Polarity and ratio test of single-phase transformers.
7. To obtain equivalent circuit, efficiency and voltage regulation of a single-phase transformer using O.C. and S.C. tests.
8. To study the construction of three phase transformer.
9. To Start DC shunt motor by using three-point starter.
10. To obtain speed control of dc shunt motor using (a) Armature resistance control (b) Field control.

Course outcomes:

Students will be able to

1. Make connection of transformer, DC Motors, DC generators, Induction motors
2. Follow safety precautions



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Semester – IV

Course Code	DEEP403
Course Title	Applied Electronics
Credits	3(L: 3 T: 0 P: 0)

Course Objective:

1. To understand the fundamental knowledge in basic electronics principles.
2. To understand ideas about the practical implementation of electronic circuits.
3. To discover design and conduct experiments in the field of electronics engineering.
4. Students will acquire new knowledge in electronics engineering discipline and to link it with electrical concepts.

Unit I

AMPLIFIERS Power Amplifiers Introduction, Classification – Class A, Class B,

Class AB & Class C, efficiency of each. Single stage class A power amplifier: Circuit operation, input & output waveforms, graphical Analysis and efficiency of Transformer couple resistive load single stage power amplifier Class A push pull amplifier Class B push pull amplifier

Class AB push pull amplifier Concept of cross over distortion Advantages of push pull amplifier collector power dissipation requirement & specifications of power transistor, need of heat sink

Unit II

MOSFET Introduction to MOSFET: Types, Construction, Working & Applications. FET Amplifier Biasing of FET: Source Self Bias, Drain to source Bias Application of FET Common Source Amplifier: Working & Applications

Unit III

FEEDBACK AMPLIFIER AND OSCILLATOR Feedback Amplifiers & oscillators General theory of feedback: Types of feedback – negative & positive feedback. Types of negative feedback – voltage shunt, voltage series, current shunt & current series. Advantages of negative feedback on voltage gain, bandwidth, input impedance output impedance, stability, noise, distortion in amplifiers. Introduction to oscillator, block diagram of sine wave oscillator, operating principles of RC & LC oscillators RC oscillators – RC phase shift, Wien bridge LC oscillators – Colpitts,



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Hartley , Crystal oscillators Circuit diagram, equation for frequency of oscillation & frequency stability.

Unit IV

Wave shaping circuits Necessity of wave shaping circuits. Linear circuits – RC integrator & differentiator – input / output waveforms & frequency response. Non-linear circuits - Clipper , diode series & shunt ,positive & negative biased & unbiased & combinational clipper.

Clampers – positive & negative clampers

Unit V

Multivibrators Transistor as switch. Definition & graphical representation of different time periods Multivibrator classification , circuit working & frequency with specific application . MMV, AMV,BMV & Schmitt trigger

COURSE OUTCOM:-

- Ability to apply knowledge of mathematics, science and engineering.
- Ability to design and conduct experiments.
- Ability to design a system, component or process to meet desired needs.
- Ability to function on multidisciplinary teams.
- Ability to identify, formulate and solve engineering problems.
- Understanding of professional and ethical responsibility.

REFERENCES

- MElectronic Principles- Paul Malvino Tata McGraw-Hill
- Applied Electronics- R.S.Sedha S.Chand & Co.
- Electronics Devices & Circuits -Allen ottershed
- Prantice Hall India LTD.
- Pulse Digital & Switching Waveforms- J.Millman and H.Taub
- Tata McGraw-Hill
- 5Pulse & Digital Electronics- G.K.Mittal and R.Vanvasai
- Khanna Publication
- Applied Electronics R.S. Sharma Foundation Publishing



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP404
Course Title	Electrical Estimation and Costing
Credits	3(L: 3 T: 0 P: 0)

Course Objective:

1. Emphasize the estimating and costing aspects of all electrical equipment, installation and designs to analyze the cost Viability.
2. Exposure to design and estimation of wiring, design of overhead and underground distribution lines, substations and illuminations design.
3. These techniques should help the students to successfully estimate costing of the products/projects that are part of our everyday usage.
4. a basic knowledge on methods and types of estimation and its merits and demerits
5. To prepare the schedule of materials with specifications and estimates for different types of electrical installations.

Unit I

Drawing and IE rules Classification of Electrical Installation. General requirement of Electrical Installation. Reading and Interpretation of Electrical Engineering Drawings. Various diagrams, plans and layout . IE rules related to Electrical Installation

Unit II

Residential Building Electrification guidelines

General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of circuits.

Unit III

Residence Building Electrification estimate and costing Method of drawing single line diagram.



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Selection of type of wiring and rating of wires & cables. Load calculations and selection of size of conductor. Selection of rating of main switch, distributions board, protective switchgear ELCB and MCB and wiring accessories. Earthing of Residential Installation. Preparation of detailed estimates and costing of Residential Installation.

Unit IV

Electrification of factory unit Installation Concept of Industrial load. Concept of Motor wiring circuit and single line diagram. Important guidelines about power wiring and Motor wiring. Design consideration of Electrical Installation in small Industry/Factory/workshop.

Unit V

Testing of Installation Testing of wiring Installation for verification of current; earthing, insulation resistance and continuity as per I

Reference books:

1.	K.B. Raina S.K. Bhattacharya	Electrical Design; Estimating and costing	New Age International (p) Limited, New Delhi
2.	Surjit Singh	Electrical Estimating and costing	Dhanpat Rai and company, New Delhi
3.	N. Alagappan S. Ekambaram	Electrical Estimating and costing	Tata Mc Graw Hill Publication, New Delhi
4.	S.L. Uappal	Electrical wiring Estimating and costing	Khanna Publication.
5.	B.D.Arora	Electrical wiring, Estimating and costing	R.B. Publication, New Delhi



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Course Outcome:-

After going through this course the student gets knowledge on:

1. The estimating and costing aspects of all electrical equipment, installation and designs to analyze the cost viability.
2. Exposure to design and estimation of wiring, design of overhead and underground distribution lines, substations and illuminations design.
3. To design illumination systems.
4. Understand types of substations and their ratings.
5. These techniques should help the students to successfully estimate costing of the products/projects that are part of our everyday usage and apply the above concepts to real-world electrical and electronics problems and applications.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP405
Course Title	Signals and Systems
Credits	3(L: 3 T: 0 P: 0)

Course Objective:-

1. To describe various signals and systems mathematically and understand how to perform mathematical operations on them.
2. Also familiar with commonly used signals such as the unit step, ramp, and impulse function, sinusoidal signals, complex exponentials and their operations.

Unit I

Classification of signals and systems: Continuous time signals (CT signals), Discrete time signals (DT signals) - Step, ramp, pulse, impulse, sinusoidal and exponential signals, basic operations on signals, classifications of CT and DT signals- Periodic and aperiodic signals, energy and power signals, random signals, CT systems and DT systems, basic properties of systems, basic properties of systems, linear time invariant systems and properties.

Unit II

Analysis of continuous time signals: Time and frequency domain analysis, Fourier series analysis, spectrum of CT signals, Fourier transform and Laplace transform, region of convergence, wavelet transform

Unit III

Linear time invariant continuous time systems: Differential equations representation, block diagram representation, state variable representation and matrix representation of systems, impulse response, step response, frequency response, realizability of systems, analog filters.

Unit IV

Analysis of discrete time signals: Convolution sum and properties, sampling of CT signals and aliasing, DTFT and properties, Z transform and properties, inverse Z transform.



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Unit V

Linear time invariant discrete time systems: Difference equations, block diagram representation, impulse response, analysis of DT LTI systems using DTFT and Z transform, state variable equations and matrix representation of systems, Digital filters.

REFERENCES:

1. Alan V. Oppenheim, Alan S. Willsky, S Hamid Nawab, 'Signals and Systems', 2nd edition 2015 Pearson New International Edition
2. A. Anand Kumar, Signals and Systems, PHI, III edition, 2015
3. Mahmood Nahvi, Signals and Systems, McGraw Hill
4. Simon Haykins and Barry Van Veen, Signals and Systems, Wiley India
5. A. Nagoor Kani; 'Signals and Systems' McGraw Hill
6. Robert A. Gabel and Richard A. Roberts, Signals & Linear Systems, Wiley.
Rodger E. Ziemer, William H. Tr

Course Outcome:-

On completion of the course, student will be able to

- 1) Analyze the discrete time signals and system using different transform domain techniques.
- 2) Design and implement LTI filters for filtering different real world signals.
- 3) Develop different signal processing applications using DSP processor.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP406
Course Title	Communication Technology
Credits	3(L: 3 T: 0 P: 0)

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Use relevant data communication technique.

Course contents:

Unit – I

Data Communication and Modulation

Block diagram of communication system

Types of communication system: synchronous and asynchronous, simplex, half-duplex, Full duplex, serial and parallel communication

Classification of communication technique: AM, FM, & PM on the basis of definition, wave-form, bandwidth, modulation index

Modulation and demodulation: Block diagram of AM, FM and PM

Pulse Modulation: Block diagram for waveform generation of PAM, PWM & PPM, working principle, advantages, disadvantages and applications.

Advantages of pulse modulation over AM and FM.

Unit – II

Digital Modulation Techniques

Digital Communication: Block diagram and working principle, waveforms, strength and limitations

Sampling process Ny-Quist sampling theorem, quantization process, quantization error, quantization noise



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Unit– III

Data Communication Media

Baud rate, Bit rate, types of errors in data communication and error correction techniques. Types of communication media and frequency band of operation

Guided media: Types of cable-twisted pair cable, co-axial cable, fiber optic cable.

Unguided media: Microwave communication, Infrared communication.

Unit– IV

Fibre Optics

- Introduction to Fiber optic communication.
- Strength and limitations of fiber optic system
- Light propagation : reflection, refraction, Snell's law
- Light propagation through cable: Mode of propagation, index profile
- Fibre optic cables: cable construction, fibre optics cable modes, single mode, step index fibre, multimode index fibre, multimode graded index fibre, fibre cable losses.
- Light source and Detector: Light emitting diode (LED), Photo Transistor, Laser diode, opto-coupler.

Unit– V

Data Communication Protocols and Interfacing Standard

OSI (Open Systems Interconnection) Reference model Introduction to protocol, FTP, SMTP, TCP/IP, UDPLAN standards.

Introduction to IEEE Standards for LAN and GPIB

RS-232 standard: Introduction, and working principle

Network topologies, introduction star, ring, tree, bus, mesh, hybrid

Basic functions of networking devices: modem, switches, routers, repeaters, hubs, bridges, gateway.

Unit-VI

Advanced Data Communication

Introduction to Wi-Fi and Wi- Max Bluetooth architecture and its layers, Universal serial bus (USB) architecture. Bluetooth and USB



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References:

1. Wayne Tomasi, Electronic Communication System, Prentice Hall of India, ISBN13:9780130494924
2. Reynders D., Steve Macky, Wright Edwin, Practical Industrial Data Communications, Newnes publication, ISBN 10:07506639523
3. George F. Kennedy, Barnard Davis, Electronic Communication System, Tata McGraw Hill, , ISBN13:9780074636824
4. Forouzan B.A., Data Communication & Networking, McGraw Hill Education; 5 edition ISBN-13: 0073376226-978

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

- a) Identify the different types of data communication equipment and techniques.
- b) Use relevant digital modulation techniques.
- c) Interpret the specifications of the data communication media.
- d) Maintain the fiber optics networks for data communication.
- e) Use OSI model and relevant data communication protocols.
- f) Maintain wireless network environment.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP407
Course Title	Illumination Practices
Credits	3(L: 3 T: 0 P: 0)

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Design illumination schemes and associated electrification of buildings.

Unit-I

Illumination Terminology Laws of Illumination 1.3 Featuring of good Illumination Scheme Advantages of good Illumination Scheme Measurement of level of Illumination (simple illumination)

Unit-II

Types of lamps: ARC lamps, HPMV lamps, Sodium Lamps, CFL Lamps, Metal halides, LED lamps Neon Sign Tubes. Neon Lamps. 2.4 Halogen Lamps. Construction, working principle, advantages, disadvantages & Application of incandescent & Florescent Lighting accessories. (All fittings, switches, encloses Illumination Auditing.

Unit-III

Purpose of lighting control Dimmer & Dimmer Transformer & their types Electronic Dimmer Enhancing Lighting control. Control circuits for lamps (refer) : ON/OFF control & Illumination control.

Unit-IV

Standard for various situations of Interior Illumination Design Techniques Design considerations for Interior location of Residential, Commercial, Industrial premises Design Illumination scheme for different Interior locations of Residential, Commercial, Industrial unit.

Unit-V

Factory Lighting Street Lighting (Latest Technology) Flood Lighting Railway Lighting ,Lighting for Advertisement/Hoardings Sports Lighting.



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References:

1. Lindsey, Jack L., Applied Illumination Engineering, The Fairmont Press Inc.
2. Simons, R. H., Bean, Robert; Lighting Engineering: Applied Calculations, Architectural Press. ISBN: 0750650516.
3. Casimer M Decusatis, Handbook of Applied Photometry, Springer, ISBN 1563964163.
4. Butterworths, Lyons Stanley, Handbook of Industrial Lighting, Butterworths
5. Simpson Robert S, Lighting Control Technology and Applications, Focal Press
6. Kao Chen, Energy Management in Illuminating Systems, CRC Press

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

- a) Select relevant lamps for various applications considering illumination levels
- b) Select the lighting accessories required for selected wiring scheme.
- c) Design relevant illumination schemes for interior applications.
- d) Design Illumination schemes for various applications
- e) Design Illumination schemes for various outdoor applications.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP408
Course Title	Utilization of Electrical Energy
Credits	3(L: 3 T: 0 P: 0)

Course Objective:

1. To clearly understand the basic concepts related to use of electric energy in various industrial, commercial and residential applications and important issues related to such usage.
2. The course also discusses issues related to power quality, economics of energy system usage and renewable energy systems.

Course Content:

Unit - I

ILLUMINATION ENGINEERING Nature of light, units, sensitivity of the eye, luminous efficiency, glare. Production of Light; Incandescent lamps, arc lamps gas discharge lamps- fluorescent lamps polar curves, effect of voltage variation on efficiency and life of lamps, Distribution and control of light, lighting calculations, solid angle, inverse square and cosine laws, methods of calculations, factory lighting, flood lighting and street lighting, Direct diffused and mixed reflection & transmission factor, refractors, light fittings.

UNIT-II

HEATING, WELDING AND ELECTROLYSIS Electrical heating-advantages, methods and applications, resistance heating, design of heating elements, efficiency and losses control. Induction heating: core type furnaces, core less furnaces and high frequency eddy current heating, dielectric heating: principle and special applications, arc furnaces: direct arc furnaces, Indirect arc furnaces, electrodes, design of heating elements, power supply and control. Different methods of electrical welding, resistance welding, arc welding, energy storage welding, laser welding, electro beam welding, and electrical equipment for them. Arc furnaces transformer and welding transformers. Review of electrolytic principles, laws of electrolysis, electroplating, anodizing-electro-cleaning, extraction of refinery metals, power supply for electrolytic process, current and energy efficiency.

UNIT-III

TRACTION Special features of Traction motors, selection of Traction Motor, Different system of electric traction and their Advantages and disadvantages, Mechanics of train movement: simplified



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speed time curves for different services, average and schedule speed, tractive effort, specific energy consumption, factors affecting specific energy consumption, acceleration and braking retardation, adhesive weight and coefficient of adhesion.

UNIT-IV

ELECTRIC DRIVES Individual and collective drives- electrical braking, plugging, rheostatic and regenerative braking load equalization use of fly wheel criteria for selection of motors for various industrial drives, calculation of electrical loads for refrigeration and air-conditioning, intermittent loading and temperature rise curve.

UNIT-V

INTRODUCTION TO ELECTRIC AND HYBRID VEHICLES Configuration and performance of electrical vehicles, traction motor characteristics, tractive effort, transmission requirement, vehicle performance and energy consumption.

RECOMMENDED BOOKS

1. Testing, Commissioning , Operation and Maintenance of Electrical Equipment by S Rao, Khanna Technical Publication, New Delhi
2. Electrical Power Systems by CL Wadhwa, Wiley Eastern Ltd., New Delhi
3. Textbook of Electrical Technology by BL Theraja, S Chand and Co., New Delhi
4. Electrical Power by Dr. SL Uppal, Khanna Publications, Delhi
5. A Course in Electrical Power by ML Soni, PV Gupta and Bhatnagar, Dhanpat Rai & Sons, New Delhi
6. Principles of Power Systems by VK Mehta, S Chand and Co., New Delhi
7. Preventive Maintenance of Electrical Apparatus by SK Sharotri, Katson Publishing House, Ludhiana

Course Outcomes:

1. To interpret the working and applications of various devices used by industry for effective utilization of electrical power.
2. To acquire the knowledge regarding the fundamentals and elementary design aspects of illumination, heating and welding.
3. To practice the fundamentals of electrolytic processes and illumination engineering



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(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP409
Course Title	Electrical Traction
Credits	3(L: 3 T: 0 P: 0)

Course Objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain electric traction systems.

Unit-1

Traction Systems and Latest Trends Explain the present scenario of Indian Railways – High speed traction, Metro 1.2 Detail the latest trends in traction- Metro, monorail, Magnetic levitation Vehicle Explain types of traction systems and their significance – Steam, diesel, diesel-electric, Battery and electric traction systems Explain the general arrangement of different types of Electric traction systems and their significance – General arrangement of D.C., A.C. single phase, 3 phase, Composite systems 1.5 Choice of traction system – Diesel - Electric or Electric.

Unit-II

.1 Draw the speed time curve related to different traction system Analysis of speed time curves for main line, suburban and urban services Solve numerical based on speed time curve - Simplified speed time curves , Relationship between principal quantities in speed time curves Calculate specific energy consumption- Requirement of tractive effort State the factors affecting Specific energy consumption - Specific energy consumption and factors affecting it.

Unit-III

State the desirable features of traction motors. Explain Significance of D.C. series motor over D.C. Shunt motor - Significance of D.C. series motor as traction motor Explain working of various A.C. motors as traction motors - A.C. Traction motors – single phase, Three phase, Linear Induction Motor Comparison between different traction motors .Apply various control methods applied to traction motors - Series – parallel control, Open circuit, Shunt and bridge transition, Pulse Width Modulation control of induction motors. Explain different types of electric braking system.



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Unit-IV

Classify electric locomotive -Important features of electric locomotives Describe the function of auxiliaries in traction system -Different types of locomotives Describe the different current collecting methods in locomotives Current collecting equipment Explain different control and auxiliary equipment used in the locomotive Describe the power conversion and transmission systems - Power conversion and transmission systems Explain Coach wiring and lighting devices- Coach wiring and lighting devices

Unit-V

explain the distribution & feeder system pertaining to traction distributions and feeders Classify traction substations - Requirements and selection Describe different methods of feeding the traction sub- station - Method of feeding the traction sub-station.

References:

1. G.C. Garg, Utilization of Electric Power & Electric Traction, Khanna Book Publishing Co., NewDelhi (ISBN: 978-93-86173-355) Revised Ed. 2018
2. Gupta J.B., S.K.Kataria and Son, Utilization of Electric power and traction
3. Partab H., Dhanpat Rai and Co,' Art and Science of Utilization of Electrical Energy
4. Partab H., Dhanpat Rai and Co, Modern Electric Traction
5. Suryanarayana N.V., New Age International Publishers, Reprint 2010
6. Open Shaw Taylor, Orient Longman ltd., Utilization of electrical energy.

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above mentioned competency:

- a) Interpret the traction layout and its systems
- b) Maintain the power supply arrangements.
- c) Maintain the function of the overhead equipment for electric traction
- d) Maintain the different components of the electric locomotive.
- e) Maintain the traction motor and train lighting system
- f) Maintain the signalling and supervisory control systems.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – IV

Course Code	DEEP410
Course Title	Electrical Testing and Commissioning
Credits	3(L: 3 T: 0 P: 0)

Course Objective:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Follow standard safety procedures in testing and commissioning of electrical equipments.

Unit – I

Electrical Safety and Insulation

Do's and don'ts regarding safety in domestic electrical appliances as well for substation/power station operators

Electrical safety in industry/power stations/ substations at the time of operation/control/maintenance. Fire detection alarm, fire-fighting equipments

Factors affecting life of insulating materials, classifications of insulating materials as per IS:1271-1958

Measuring insulation resistance by different methods such as i) Polarization, ii) Dielectric absorption, iii) Megger and to predict the condition of insulation

Reconditioning of insulation,

Insulating oil - properties of insulating oil, causes of deterioration of oil, testing of transformer oil as per IS 1866-1961

Unit – II

Installation and Erection

Concept of foundation for installation of machinery. Requirements of foundation for static and rotating electrical machinery.

Concept of leveling and aligning Procedure for leveling and aligning alignment of direct coupled drive, effects of mis-alignment



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Installation of transformer as per I.S.-1886-1967 and procedure of installation of transformer, Requirements of installation of pole mounted transformer

Requirements of installation of rotating electrical machines as per I.S. 900 - 1965 Devices and tools required for loading, unloading, lifting, and carrying heavy equipment and precautions to be taken while handling them.

Unit– III

Testing and Commissioning

Concept of testing, Objectives of testing. Roles of I.S.S. in testing of electrical equipment, Types of tests and concepts, Routine tests, type tests, supplementary test, special tests, Methods of testing - Direct/Indirect/Regenerative testing.

Tolerances for the various items for equipment –transformer, induction motor, dc motor, synchronous machines

Commissioning, Tests before Commissioning for transformer, induction motor,

Unit- IV

Troubleshooting Plans

Internal and external causes for failure / abnormal operation of equipment.

List of mechanical faults, electrical faults and magnetic faults in the electrical equipment remedies, applications

Use of tools like bearing puller filler gauges, dial indicator, spirit level, megger, earth tester, and growler. Common troubles in electrical equipments and machines.

Preparation of trouble shooting charts for D.C. Machines, AC Machines and transformers.

Unit– V

Maintenance

Concept of maintenance, types of maintenance, Routine, preventive and breakdown maintenance.

Causes of failure of electrical machines

Preventive maintenance-procedure or developing maintenance schedules for electrical machines.

Factors affecting preventive maintenance schedules, Concept of TPM, Pillars of TPM

Identification of different types of faults developed such as mechanical/ electrical/ magnetic

Faults Maintenance schedules of the following as per I.S.S.



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- a) Distribution transformer as per I.S.1886-1967
- b) Single phase and three phase Induction motors as per I.S.900-1965.
- c) Batteries

References:

1. Deshpande.M. V. PHI Learning Pvt. Ltd., 2010, Design and Testing of Electrical Machines ISBN No8120336453, 9788120336452.
2. Rao, B V S Asia Club House, First Reprint, 2011, Operation and Maintenance of Electrical Equip-ment Vol-I, ISBN No 8185099022
3. Rosenberg. Mc GRAW-HILL, 1st Edition, May 2003, Maintenance and Repairs, ISBN No 9780071396035
4. Sharotri, S.K. Glencoe/ Mcgraw- Hill; 2ndEdition , June 1969; Preventive Maintenance of Electrical Apparatus, ISBN No 10: 007030839X 13: 978-0070308398

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

- a) Follow safety procedures with respect to earthing and insulation of electrical equipment
- b) Select proper tools, equipment, for installation, testing, maintenance of electrical machines and transformers
- c) Test and commission electrical equipment in accordance with IS codes
- d) Make plans for troubleshooting electrical machines.
- e) Undertake regular preventive and breakdown maintenance.



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Credit System and Marks Distribution:-

Semester-V												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	PCC	DEEP501	Power System -II	3	0	0	3	30	70	21	100	35
2	PCC	DEEP502	Electrical Machine -II	3	0	0	3	30	70	21	100	35
3	PCC	DEEP503	Microprocessor & Microcontroller	2	0	0	2	30	70	21	100	35
4	PEC	DEEP(E)	Program Elective-III	3	0	0	3	30	70	21	100	35
5	PEC	DEEP(E)	Program Elective-IV	3	0	0	3	30	70	21	100	35
6	OEC	DEEO(E)	Open Elective-I	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	PCC	DEEP551	Power system II Lab	0	0	2	1				50	25
2	PCC	DEEP552	Electrical Machine -II Lab	0	0	2	1				50	25
	PCC	DEEP553	Microprocessor & Microcontroller Lab	0	0	2	1					
3	IN	DE554	Summer Internship-II (after 4 th sem)	0	0	0	3				50	25
TOTAL							23					



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Elective Papers for semester-V

Program Elective-III (Choose any one)												
SL No	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	PEC	DEEP504(E)	Switch gear and Protection	3	0	0	3	30	70	21	100	35
2	PEC	DEEP505(E)	Wind and Solar energy System	3	0	0	3	30	70	21	100	35
3	PEC	DEEP506(E)	Electric Drives	3	0	0	3	30	70	21	100	35
Program Elective-IV (Choose any one)												
1	PEC	DEEP507(E)	Industrial Instrumentation	3	0	0	3	30	70	21	100	35
2	PEC	DEEP508 (E)	Solar Technology	3	0	0	3	30	70	21	100	35
3	PEC	DEEP509(E)	biomedical	3	0	0	3	30	70	21	100	35
Open Elective-I (Choose any one)												
1	OEC	DEEO510(E)	Solid Waste Management	3	0	0	3	30	70	21	100	35
2	OEC	DEEO511(E)	Power Plant Engineering	3	0	0	3	30	70	21	100	35
3	OEC	DEEO512(E)	Web Technology	3	0	0	3	30	70	21	100	35



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma (EEE) Semester – V

Course Code	DEEP501
Course Title	Power System -II
Credits	3(L: 3 T: 0 P: 0)

Course objectives:

1. Understand interconnected power systems
2. Understand various method of power flow study
3. Understand the frequency control
4. Analyze power system stability.
5. To understand the concept of power flow mathematically

Unit I

Transients in Power System and protection against them. Elements of Power System dynamics, Computer methods in Power System Analyses, load Flow Studies.

Unit II

Power transmission systems – Electrical characteristics of overhead lines and cables, Sag & Tension, Proximity, Corona, Skin effect, Bundled conductors, Transposition of conductors, Per unit representation of system quantities. Steady state performance of transmission network – ABCD parameters of short, medium and long lines.

Unit III

Methods of active and reactive power control – use of VAR compensators Elements of economic operations. Electricity tariffs. Distribution systems – feeders and distributors radial and mesh system – distribution sub-station – earthing, Important I.E. rules Indian Electricity Acts.

Unit IV

Nature of faults in electrical systems, fault calculations, symmetric and asymmetric faults – method of sequence components – sequence networks.

Unit V

Study of Protective Devices- Isolators, disconnecting switch, lightening arrester, Horn gap, CT, PT, Protective relays and their applications to power apparatus and systems. Principles of circuit breakers – different types, oil circuit breakers, air circuit breakers, vacuum circuit breakers, SF₆ – circuit breakers, their uses and comparison.



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REFERENCES:

1. Stevenson W.D. : Elements of Power System Analysis; McGraw Hill International Edn.
2. Grainger J J & Stevenson W.D. Power System Analysis ; McGraw Hill Intl. Edn.
3. Nagrah I J & Kothari D.P. : Power System Engg. TMH Pub. Co. Ltd.

Courses outcomes: students will be able to

1. Analyze interconnected power systems
2. Analyze various method of power flow study
3. Analyze the frequency control methods
4. Analyze power system stability



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester-V

Course Code	DEEP551
Course Title	Power Systems-II Lab
Credits	1(L: 0 T:0 P: 2)

Course objective:

To understand practically faults, transmission line parameters, CT /PT relays etc.

List of Experiments:

1. To verify various faults in transmission lines.
2. To measure the electrical power
3. Study of phase lag and phase lead.
4. Measurement of transmission line Inductance and capacitance
5. Working of CT and PT.

Course Outcomes:

Students will be able

1. To analyze fault types in transmission line can measure transmission line parameters
2. They know the use of relay ,CT and PT



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEP502
Course Title	Electrical Machine -II
Credits	3(L: 3 T: 0 P: 0)

Courses objectives: students will be able to

1. Understand construction and working principle of 3 phase induction motor.
2. Understand starting and speed control methods.
3. Understand the construction and working principle of synchronous motor
4. Analyze characteristics of synchronous motor and its speed control

Unit -I

Construction of three phase induction motor Production of rotating magnetic field Principle of working/operation 1.4 Concept of slip. Equation of rotor induced emf, current, frequency, reactance, and impedance under steady and running condition Torque equation of three phase induction motor 1.7 Starting and running torque of squirrel cage and slip ring induction motor Condition for maximum and starting torque 1.9 Torque slip characteristics of three phase induction motor 1.10 Effect of change in rotor circuit resistance on torque-slip characteristics Effect of change in supply voltage on torque-slip characteristics 08 1.12 measurement of slip by a) Tachometer method b) Comparing rotor frequency and stator frequency 1.13 Speed control of three phase induction motor by a) Pole changing method b) Frequency control method c) By stator voltage control d) Rotor resistance control

Unit-II

Definition and construction of three phase Alternator a) Armature b) Rotor- smooth cylindrical & projecting type Derivation of e.m.f. equation of Alternator which includes a) Chording factor b) Distribution factor Factors affecting the terminal voltage of Alternator a) Armature resistive drop b) Leakage reactance drop c) Armature reaction at various power factors & concept of Synchronous impedance .Regulation of three phase Alternator by a) Synchronous impedance method b) mmf method



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Unit-III

Synchronous Motor 3.1 Principle of working/operation 3.2 Synchronous Motor on load with constant excitation 3.3 Effect of excitation at constant load 3.4 V curve & inverted V curve 3.5 Hunting & phase swinging 3.6 Applications 3.7 Starting of Synchronous Motor 3.8 Comparison between IM & Synchronous Motor (Numerical on all above) Single phase Motors Double field revolving theory Types of Single phase IM 4.3 Split phasing principle of starting a) Resistance start induction run b) Capacitor start induction run c) Capacitor start Capacitor run d) Double value Capacitor applications.

Unit-IV

Shaded pole IM Applications Special machines Induction Generator: Principle of operation, Construction and Applications Linear Induction Motor Principle of operation, Construction and Applications AC series motor Principle of operation, Construction and Applications

BOOKS

1. Electrical Machines by SK Bhattacharya, Tata Mc Graw Hill, New Delhi
2. Electrical Machines by SK Sahdev, Unique International Publications, Jalandhar
3. Electrical Machines by Nagrath and Kothari, Tata Mc Graw Hill, New Delhi
4. Electrical Machines by SB Gupta, SK Kataria and Sons, New Delhi

Courses outcomes: students will be able to

1. Understand working principle of 3 phase induction motor.
2. Understand starting and speed control methods.
3. Understand the working principle of synchronous motor
4. Analyze characteristics of synchronous motor.



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Course Code	DEEP 552
Course Title	Electrical Machine –II lab
Credits	1(L: 0 T: 0 P: 2)

Courses objectives:

1. Understanding starting and speed control methods of induction motor practically.
2. Understanding starting and speed control methods of synchronous motor practically
3. Understanding the characteristics of BLDC Motors
4. Understanding the characteristics of Reluctance motor.
5. Understanding the characteristics of Stepper motor.

List of Experiments Expandable:

1. Starting methods of Single -phase Induction motor
2. Speed Control of single- phase induction motor
3. Starting and speed control of synchronous motor
4. Starting methods of Three -phase Induction motor
5. Speed Control of Three - Phase induction motor
6. To study the characteristics of BLDC Motors
7. To study the characteristics of Reluctance motor.
8. To study the characteristics of Stepper motor.

Courses outcomes: students will be able to

1. Make connection for various experiments on induction motor, Synchronous motor, BLDC and stepper motor.
2. Follow safety precautions



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING (Diploma EEE)

Semester – V

Course Code	DEEP 503
Course Title	Microprocessor & Microcontroller
Credits	2(L: 2 T: 0 P: 0)

Course Objective:

1. To provide solid foundation on the fundamentals of microprocessors and applications,
2. To understanding the interfacing, the external devices to the processor according to the user requirements thus, enabling to create novel products and solutions for real time problems

MODULE – I

Digital computer, Computer languages, Main frame, Mini computers, Microcomputers, Architecture of 8085 microprocessors, Functions of different pins, Bus Concept.

MODULE – II

Memory organization, Memory map, Interfacing devices, Memory interfacing, Different machine cycles

MODULE – III

Instruction set, Instruction classification, Instruction format, Addressing modes of 8085, Simple illustrative programs and flow chart, System timing diagram.

MODULE – IV

Programming techniques, Looping, Counting, Logic operations, Sorting, Counter and time delays, Stack and subroutine, Code conversion BCD to binary, Binary to BCD, Binary to ASCII and ASCII to Binary, BCD Arithmetic.

Data transfer schemes, Memory mapped I/O and I/O mapping, I/O port Intel 8212 interfacing with multiplexed 7-segment LED and matrix keyboard, Intel 8255 all modes, Timer 8253/8254 Keyboard/Display Interface 8279, Control words and interfacing.

MODULE – V

Interrupt structure of 8085, Hardware and software interrupts, EI, DI, RIM and SIM instructions, Interfacing DAC 1408 and staircase ramp and triangular wave form generation, Interfacing ADC 0801, Applications.

MODULE – VI

Introduction to microcontroller, Popular microcontroller, Applications, Architecture of 8051 microcomputers, Internal and external memories, Interrupts.



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Course Outcomes:

At the end of this course students will be able to:

1. Understand the fundamentals of Microprocessors.
2. Understand the internal design of 8051 microcontrollers along with the features and their programming.

Text Books:

1. Ramesh S. Gaonkar, Microprocessor Architecture - Programming, Applications
2. Raj Kamal, Microcontrollers - Architecture, Programming, Interfacing and System Design, Pearson Education.

Reference Books:

1. Renu Singh and B. P. Singh, Microprocessors, Interfacing and Applications, New Age International Publication.
2. A.P. Malvino, Digital Computer Electronics
3. S. K. Venkatram, Advanced Microprocessor & Microcontroller
4. A. P. Mathur, Introduction to Microprocessors



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEP 553
Course Title	Microprocessor & Microcontroller Lab
Credits	1(L: 0 T: 0 P: 2)

Course objectives:

1. To understand the ALP for 8085/8051 using trainer kit, Keil compiler etc
2. To interface various hardware using 8085/8086/8051 etc

List of experiments (expandable)

1. Store data in memory location and registers in 8085
2. Store data in memory location and registers in 8085
3. ALP of addition of two numbers in 8085/8086/8051
4. ALP of addition of two numbers in 8085/8086/8051
5. Programming using Kiel C51 software

Course outcomes: students will be able to develop ALP and execute them on trainer kit or Keil compiler etc.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEP504
Course Title	Switch Gear and Protection
Credits	3(L: 3 T: 0 P: 0)

Course Objective:

1. Explain the working of different types of switchgear equipment like circuit breakers and relays.
2. Design the ratings for fuses according to the requirement
3. Elucidate various protection schemes of various power system components like alternators, transformers and bus-bars.
4. Explain various methods of over voltage protection in power systems.

UNIT I

Fundamental: Necessity & functions of protective system. Normal & abnormal conditions. Types of faults & their causes. Short circuit calculations (Symmetrical faults only). Use of current limiting reactors & their arrangements. **Circuit interrupting devices:** Fuses-Construction, Working of Semien closed & HRC fuse, characteristics, selection and applications. Isolators-Vertical break, Horizontal break & Pento graph type. Arc formation process, methods of arc extinction – High resistance method, Low resistance or current zero method, related terms–Arc voltage, Recovery voltage & Re striking voltage.

Circuit breakers -Concept, Classification, Working principle, Construction, Specification & Applications of H.T–Bulk oil circuit breaker, Minimum oil circuit breaker (M.O.C.B.), Sulphur Hexa Fluoride circuit breaker (SF₆), Vacuum circuit breaker. L.T.-Air circuit breakers (ACB), Miniature circuit. Breakers (MCB), Moulded case circuit breaker (MCCB), Earth leakage circuit breaker (ELCB), Comparison of use & MCCB. Selection of MCCB for motor. Selection and rating of circuit breakers.

UNIT II

Protective Relaying: Fundamental Requirements (Qualities)-Selectivity, Speed, Sensitivity, Reliability, Simplicity, Economy. Basic relay terminology-Protective relay, relay time, Pick up current, Reset current, Current setting, Plug setting multiplier, Time setting multiplier. Classification Electromagnetic attraction– Operation of Attracted armature type, Solenoid type and Balanced beam type relays. Electromagnetic induction type–Operation of Shaded pole type and



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Watt hour meter type relays. Block diagram, Operation, Advantages & disadvantages of Static and μ P based relays.

UNIT III

Differential Relay- Operation of Current differential relay & Voltage differential relay. Protection of Alternator Abnormalities & Faults, Differential protection, Differential protection, over current, earth fault, inter turn fault, negative phase sequence, over heating protection. Reverse power protections. **Protection of** Abnormalities & faults. Differential, Biased differential, Over current, Earth fault, Inter turn, Restricted earth fault, Over heating protection. Buchholz relay. Protection of Motor: Abnormalities & faults. Short circuit protection, Overload protection, Single phase preventor.

UNIT IV

Protection of Bus bar & transmission line : Abnormalities & faults. Bus bar protection – Operation of Differential Protection and Fault bus protection schemes. Transmission line, over current, distance protection. Pilot wire protection. **Neutral Earthing :** Introduction & importance. Types of earthing. Substation earthing. Difference between Equipment earthing and Neutral earthing. **Over voltage Protection :** Causes of over voltages. Lightning phenomena & over voltage due to lightning .Protection of transmission line & substation from direct stroke. Types of lightning arresters – Rod gap, Horn gap, Expulsion and Thyristor type , their construction & principle of operation. Surge absorber – Definition & working with neat diagram. Protection against traveling waves. Necessity of Insulation co-ordination.

Suggested Reading:

1. Power System Protection & Switch Gear : Badrinarayana and Vishwa Karma, TMH Publications.
2. Switch Gear and Protection Sunil S. Rao, Khanna Publications

Course Outcomes:

1. To understand the types of Circuit breakers and relays for protection of Generators, Transformers and feeder bus bar from Over voltages.
2. To describe the importance of neutral grounding for overall protection.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEP505
Course Title	Wind and Solar Energy systems
Credits	3(L: 3 T: 0 P: 0)

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

1. Maintain large wind power plants and small wind turbines
2. Maintain the efficient operation of various types of solar power technologies

Module 1: Physics of Wind Power:

History of wind power, Indian and Global statistics, Wind physics, Betz limit, Tip speed ratio, stall and pitch control, Wind speed statistics-probability distributions, Wind speed and power-cumulative distribution functions.

Module 2: Wind generator topologies:

Review of modern wind turbine technologies, Fixed and Variable speed wind turbines, Induction Generators, Doubly-Fed Induction Generators and their characteristics, Permanent- Magnet Synchronous Generators, Power electronics converters. Generator-Converter configurations, Converter Control.

Module 3: The Solar Resource

Introduction, solar radiation spectra, solar geometry, Earth Sun angles, observer Sun angles, solar day length, Estimation of solar energy availability.

Module 4: Solar photovoltaic:

Technologies-Amorphous, monocrystalline, polycrystalline; V-I characteristics of a PV cell, PV module, array, Power Electronic Converters for Solar Systems, Maximum Power Point Tracking (MPPT) algorithms.Converter Control.

Module 5: Network Integration Issues:

Overview of grid code technical requirements. Fault ride-through for wind farms - real and reactive power regulation, voltage and frequency operating limits, solar PV and wind farm behavior during grid disturbances. Power quality issues. Power system interconnection experiences in the world.



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Hybrid and isolated operations of solar PV and wind systems.

Module 6: Solar thermal power generation:

Technologies, Parabolic trough, central receivers, parabolic dish, Fresnel, solar pond, elementary analysis.

Text / References:

1. T. Ackermann, “Wind Power in Power Systems”, John Wiley and Sons Ltd., 2005.
2. G. M. Masters, “Renewable and Efficient Electric Power Systems”, John Wiley and Sons, 2004.
3. S. P. Sukhatme, “Solar Energy: Principles of Thermal Collection and Storage”, McGraw Hill, 1984.
4. H. Siegfried and R. Waddington, “Grid integration of wind energy conversion systems” John Wiley and Sons Ltd., 2006.
5. G. N. Tiwari and M. K. Ghosal, “Renewable Energy Applications”, Narosa Publications, 2004.
6. J. A. Duffie and W. A. Beckman, “Solar Engineering of Thermal Processes”, John Wiley & Sons, 1991



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEP506
Course Title	Electric Drives
Credits	3 (L: 3, T:0 P: 0)

Course Objective:

At the end of this course, students will demonstrate the ability to

- Understand the characteristics of dc motors and induction motors.
- Understand the principles of speed-control of dc motors and induction motors.
- Understand the power electronic converters used for dc motor and induction motor speed control.

UNIT 1 DC motor characteristics

Review of emf and torque equations of DC machine, review of torque-speed characteristics of separately excited dc motor, change in torque-speed curve with armature voltage, example load torque-speed characteristics, operating point, armature voltage control for varying motor speed, flux weakening for high-speed operation.

UNIT 2: Multi-quadrant DC drive

Review of motoring and generating modes operation of a separately excited dc machine, four quadrant operation of dc machine; single-quadrant, two-quadrant and four-quadrant choppers; steady-state operation of multi-quadrant chopper fed dc drive, regenerative braking.

UNIT 3: Chopper fed DC drive

Review of dc chopper and duty ratio control, chopper fed dc motor for speed control, steady state operation of a chopper fed drive, armature current waveform and ripple, calculation of losses in dc motor and chopper, efficiency of dc drive, smooth starting.

UNIT 4: Closed-loop control of DC Drive

Control structure of DC drive, inner current loop and outer speed loop, dynamic model of dc motor – dynamic equations and transfer functions, modeling of chopper as gain with switching delay, plant transfer function, for controller design, current controller specification and design, speed controller specification and design.



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UNIT 5: Induction motor characteristics

Review of induction motor equivalent circuit and torque-speed characteristic, variation of torque-speed curve with (i) applied voltage, (ii) applied frequency and (iii) applied voltage and frequency, typical torque-speed curves of fan and pump loads, operating point, constant flux operation, flux weakening operation.

Review of three-phase voltage source inverter, generation of three-phase PWM signals, sinusoidal modulation, space vector theory, conventional space vector modulation; constant V/f control of induction motor, steady-state performance analysis based on equivalent circuit, speed drop with loading, slip regulation

Control of slip ring induction motor

Impact of rotor resistance of the induction motor torque-speed curve, operation of slip-ring induction motor with external rotor resistance, starting torque, power electronic based rotor side control of slip ring motor, slip power recovery.

Text / References:

1. G. K. Dubey, "Power Semiconductor Controlled Drives", Prentice Hall, 1989.
2. R. Krishnan, "Electric Motor Drives: Modeling, Analysis and Control", Prentice Hall, 2001.
3. G. K. Dubey, "Fundamentals of Electrical Drives", CRC Press, 2002.
4. W. Leonhard, "Control of Electric Drives", Springer Science & Business Media, 200

Course Outcomes:

At the end of this course, students will demonstrate the ability to

- Understand the characteristics of dc motors and induction motors.
- Understand the principles of speed-control of dc motors and induction motors.
- Understand the power electronic converters used for dc motor and induction motor speed control.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEP507
Course Title	INDUSTRIAL INSTRUMENTATION
Credits	3 (L: 3, T:0 P: 0)

COURSE OBJECTIVES:

1. To introduce the measurement techniques of force, torque and speed.
2. To introduce the measurement techniques of acceleration, Vibration and density
3. To introduce the measurement Viscosity, Humidity and moisture.
4. To introduce the temperature measurement techniques
5. To introduce the pressure measurement techniques

UNIT I MEASUREMENT OF FORCE, TORQUE AND SPEED

Load Cell introduction, Different types of load cells: Hydraulic, Pneumatic, Strain gauge, Magneto-elastic and Piezoelectric load cells – Different methods of torque measurement: Strain gauge, Relative angular twist. Speed measurement: Capacitive tacho, Drag cup type tacho, D.C and A.C tacho generators – Stroboscope.

UNIT II MEASUREMENT OF ACCELERATION, VIBRATION AND DENSITY

Accelerometers: LVDT, Piezoelectric, Strain gauge and Variable reluctance type accelerometers – Mechanical type vibration instruments – Seismic instruments as accelerometer – Vibration sensor – Units of density and specific gravity: Pressure type densitometers, Float type densitometers, Ultrasonic densitometer and gas densitometer.

UNIT III MEASUREMENT OF VISCOSITY, HUMIDITY AND MOISTURE

Viscosity: Say bolt viscometer – Rotameter type and Torque type viscometers – Consistency Meters – Humidity: Dry and wet bulb psychrometers – Resistive and capacitive type hygrometers – Dew cell – Commercial type dew meter. Moisture: Different methods of moisture measurements – Thermal, Conductivity and Capacitive sensors, Microwave, IR and NMR sensors, Application of moisture measurement – Moisture measurement in solids.



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UNIT IV TEMPERATURE MEASUREMENT

Definitions and standards – Primary and secondary fixed points – Different types of filled in system thermometers – Sources of errors in filled in systems and their compensation – Bimetallic thermometers – IC sensors – Thermocouples: Laws of thermocouple, Fabrication of industrial thermocouples, Reference junctions compensation, – Radiation fundamentals – Radiation methods of temperature measurement – Total radiation pyrometers – Optical pyrometers – Two colour radiation pyrometers – Fibre optic sensor for temperature measurement – Thermograph, Temperature switches and thermostats – Temperature sensor selection, Installation and Calibration.

UNIT V PRESSURE MEASUREMENT

Units of pressure – Manometers: Different types, Elastic type pressure gauges: Bourdon tube, Bellows, Diaphragms and Capsules – Electrical methods: Elastic elements with LVDT and strain gauges – Capacitive type pressure gauge – Piezo resistive pressure sensor-Resonator pressure sensor – Measurement of vacuum: McLeod gauge, Thermal conductivity gauge, ionization gauges, Cold cathode type and hot cathode type – Pressure gauge selection, installation and calibration using dead weight ester.

Course outcomes:

After studying this course students will be able to

1. Describe the basic principle of transducers
2. Applications of transducers in real world
3. Measurement, calibration and signal conditioning of transducers.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEP508
Course Title	Solar Technology
Credits	3 (L: 3, T:0 P: 0)

Course Objective: To explain the technical and physical principles of solar cells and solar collectors, measure and evaluate different solar energy technologies through knowledge of the physical function of the devices, calculate the required size of solar cell systems and solar collectors from a given power need and socio-economic issues around solar energy in a concise and an accessible way to a target group with basic technical skills.

Unit – I

Solar Radiation, Solar Cells and Modules: Properties of sunlight. Absorption by the atmosphere. Calculation of solar irradiance at surfaces. The function of solar cells from semiconductor physics. Different solar cell technologies and fabrication methods. Concepts for increasing efficiency based on loss analysis. Wavelength sensitivity. Series connection of solar cells to modules. Module function and characteristics. Shading of cells and modules

Unit – II

Solar Cell Systems System components and their functions. Calculating output and dimensioning of solar cell systems. Analysis and computer simulation of a solar cell system. Concentrated sunlight and solar power (CSP). Properties of optical concentration systems. Solar cells in concentrated sunlight. Overview of the different components in a CSP system and their functions. Examples of CSP-systems globally.

Unit – III

Solar thermal Thermodynamic description of solar collectors. Optical properties of solar collectors and technologies for fabrication. Solar thermal systems for different applications in Sweden and abroad. Storage of solar generated heat.

Unit – IV

Hybrid systems Combinations of solar thermal and solar cell systems. Overview of different applications. District heating with solar thermal components.



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(Diploma EEE)

Unit – V

Active solar energy in systems: How large-scale deployment of active solar energy is possible in Sweden and globally. Buying and selling heat and electric energy. Grid aspects of large-scale deployment of solar cells as well as environmental and socioeconomic aspects.

Books Recommended:

1. A Text book of Power System Engineering, A Chakrabarti, M. L. Soni, P. V. Gupta, U. S. Bhatnagar, Dhanpat Rai Publication
2. Renewable Energy Technologies, Solanki, Chetan S. , PHI Learning, New Delhi, 2011
3. Renewable Energy Sources for Sustainable Development, N.S. Rathore and N. L. Panwar, New India Publishing Agency, New Delhi
4. Renewable Energy Resources, J. Twidell and T. Weir, E & F N Spon Ltd, London, 1999
5. Electric Power Generation: Transmission and Distribution, S. N. Singh, PHI Learning, New 6. Electrical Power, Dr. S.L. Uppal

Course Outcome:

- State the different types of Solar Cells, Modules and Solar Radiation, Explain the Need, importance and scope of Solar Energy resources.
- Describe the working System components and their functions. Calculating output and dimensioning of solar cell systems. Explain Examples of CSP-systems globally.
- Describe the working of Solar Thermal Power Plants using single line diagram and state the functions of the major equipment and auxiliaries of each plant.
- Describe the working of Prepare economic analysis for Commercial/ Industrial/ Residential PV Hybrid energy conservation systems



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEP509
Course Title	Bio-medical
Credits	3 (L: 3, T:0 P: 0)

Course objectives:

The course has the following objectives:

- To introduce an fundamentals of transducers as applicable to physiology
- To explore the human body parameter measurements setups
- To make the students understand the basic concepts of forensic techniques.

Unit 1

Cell resting potential and action potentials – Origin of bio potentials – characteristics – Frequency and amplitude ranges – ECG – Einthoven's triangle – 3 lead ECG system – EEG – 10- 20 electrode system – Origin and characteristics of EMG – EOG – ERG electrodes and transducers.

Unit 2

Electrode-electrolyte interface – Electrode – skin interface – Half cell potential – Impedance – Polarization effects of electrode – Nonpolarizable electrodes. Types of electrodes – Surface; needle and micro electrodes – ECG – EMG – EEG Electrodes.

Unit 3

Diagnostic and Therapeutic Equipments: Blood pressure monitors – Electrocardioscope – Pulse Oximeter – pH meter – Auto analyzer – Pacemakers – Defibrillator – Heart lung machine – Nerve and muscle stimulators – Dialysis machines – Surgical diathermy equipments – Nebulizer; inhalator – Aspirator – Humidifier – Ventilator and spirometry.

Unit 4

Medical imaging techniques: Basics of diagnostic radiology – Production – Nature and properties of X rays – X-ray machine – Block diagram – Digital radiography – CT – Basic Principle – Block diagram – Radioisotopes in medical diagnosis – Physics of radioactivity – Gamma Camera. Block diagram – SPECT Scanner – PET Scanner – Principles of NMR Imaging systems – Block diagram



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of NMR Imaging System – Ultrasonic Imaging Systems – Physics of Ultrasound waves – Doppler effect – Medical Ultrasound Electrical safety: Physiological effects of electricity. Micro & macro shock hazards – Electrical Safety codes and standards – Protection of patients.

Text Book

- R S Khandpur, “Handbook of Biomedical Instrumentation”, 1st ed., Tata McGraw Hill Publishing Company Limited, 2004.References
- John G Webster, “Medical Instrumentation – Application and Design”, 4th ed., John Wiley and Sons, 2007.
- Leslie Cromwell, Fred. J. Weibell, Erich. A. Pfeiffer, “Biomedical Instrumentation & Measurements, 2nd ed., Pearson Education., 2001.

Learning Outcomes: After the successful completion of the course the students will be able to:

1. Differentiate and analyses the biomedical signal sources.
2. Elucidate cardiovascular system and related measurements.

Course Outcome:

1. Recollect the basic knowledge of anatomy and physiology of human body.
2. Understand & Analyze important vital sign parameters to evaluate certain disease.



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(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEO511
Course Title	Power Plant Engineering
Credits	3 (L: 3, T:0 P: 0)

Course Objective:

- To introduce students to different aspects of power plant engineering.
- To familiarize the students to the working of power plants based on different fuels.
- To expose the students to the principles of safety and environmental issues.

UNIT I

INTRODUCTION & ECONOMICS OF POWER PLANT

Power plant-Introduction, Classification - Location of power plant- Choice of Power plant Terminology used in power plant: Peak load, Base load, Load factor, Load curve, demand factor- Various factor affecting the operation of power plant- Load sharing- cost of power tariff methods- factors involved in fixing of a tariff.

UNIT II

HYDRO POWER PLANT

Hydro electric power plant- Introduction, storage and poundage, Selection of sites for hydro electric power plant-General layout and essential elements of Hydro electric power plant and its working- Classification of the plant- base load plant, peak load plant, Run off river plant, storage river plant, pumped storage plant, mini and micro hydel plants, governing of hydraulic turbines- impulse turbine-reaction turbine, selection of turbines, Advantages and disadvantages-limitations of hydro electric power plant.

UNIT III

THERMAL POWER PLANT

Thermal power plant -General layout, working, Site Selection, materials required for thermal power plants, coal handling and its methods, stages in coal storage, Fuel burning-Stoker firing, overfeed stoker, under feed stokers, chain grate stoker, Pulverized fuel handling system-unit and central system, Pulverization of coal-Ball mill, Ash handling system, Gravity system, electrostatic precipitation (ESP) system, Feed water treatment- Mechanical method, Advantages and



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disadvantages and limitations of Thermal power plant.

UNIT IV

NUCLEAR POWER PLANT & GAS TURBINE POWER PLANTS

Nuclear power plant-introduction-nuclear fuels, nuclear fission and fusion, working of a nuclear power plant, types of reactors- pressurized water reactor- boiling water reactor, effects of nuclear radiation, different methods for nuclear waste disposal-low, medium and high level waste disposal, Advantages -disadvantages- limitations. Gas turbine power plant- Schematic diagram & working of open and closed cycle gas turbine power plant, Components of Gas turbine-compressor, combustion chamber, gas turbine, gas turbine fuels, Advantages -disadvantages- limitations of Gas turbine powerplant

UNIT V

SOLAR, WIND AND DIESEL POWER PLANTS.

Solar power plant-introduction, layout, Solar cell fundamentals & classification – maximum power point tracker (MPPT) and solar panel.

Wind power plant: introduction, Factors affecting distribution of Wind energy, Variation of wind speed with height and time-Horizontal axis wind turbine (HAWT)-types of rotors Vertical axis wind turbine- types of rotors- Wind energy conversion system (WECS) advantages and disadvantages-limitations of Wind power plant.

Diesel power plant- layout, Components and working- Advantages –disadvantages, limitations.

UNIT VI

PLANT SAFETY AND ENVIRONMENTAL IMPACT OF POWER PLANT

Social and Economical issues of power plant- Oxides of sulphur, oxides of carbon, oxides of nitrogen, Acid precipitation-Acid rain- acid snow- Dry deposition-acid fog, green house effect, air and water pollution from thermal power plants and its control, Thermal pollution from thermal power plants, noise pollution and its control, natural and artificial radioactivity nuclear power and environment- radiations from nuclear power plant effluents- high level wastes- methods to reduce pollution, global warming- its effects and control, standardization for environmental pollution.



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Reference Books:

1. Power Plant Engineering” Mahesh Verma, Metropolitan Book Company Pvt. Ltd.New Delhi
2. “Power Plant Technology” El-Vakil, McGraw Hill.
3. Power Plant Engineering by P.K. Nag, Tata McGraw Hill.
4. Steam & Gas Turbines & Power Plant Engineering by R.Yadav, Central Pub.House.

Course Outcomes:

At the end of the course, a student will be able to:

1. Describe and analyze different types of sources and mathematical expressions related to thermodynamics and various terms and factors involved with power plant operation.
2. Analyze the working and layout of steam power plants and the different systems comprising the plant and discuss about its economic and safety impacts
3. Combine concepts of previously learnt courses to define the working principle of diesel power plant, its layout, safety principles and compare it with plants of other types.
4. Describe the working principle and basic components of the nuclear power plant and the economic and safety principles involved with it.
5. Discuss the working principle and basic components of the hydroelectric plants and the economic principles and safety precautions involved with it



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEO510
Course Title	Solid Waste Management
Credits	3 (L: 3, T:0 P: 0)

Course Objectives:

Following are the objectives of this course:

- To know various sources of solid.
- To learn techniques of collection and transportation of solid waste.
- To know various methods of disposal of solid waste.
- To understand and identify different biomedical and E-waste and their subsequent disposal techniques.

Course Content:

Unit – I Introduction

- Definition of solid waste, different solid waste – domestic Waste, commercial waste, industrial waste, market waste, agricultural waste, biomedical waste, E-waste, hazardous waste,
- institutional waste, etc.
- Sources of solid waste, Classification of solid waste – hazardous and non- hazardous waste.
- Physical and chemical characteristics of municipal solid waste.

Unit– II Storage, Collection and Transportation of Municipal Solid Waste

- Collection, segregation, storage and transportation of solid waste.
- Tools and Equipment-Litter Bin, Broom, Shovels, Handcarts, Mechanical road sweepers,
- Community bin - like movable and stationary bin.
- Transportation vehicles with their working capacity -Animal carts, Auto vehicles, Tractors, Trailers, Trucks, Dumpers, Compactor vehicles. Transfer station- meaning, necessity, location.
- Role of rag pickers and their utility for society.

Unit– III Composting of Solid Waste

1. Concept of composting of waste, Principles of composting process. Factors affecting the composting process.
2. Methods of composting – Manual Composting – Bangalore method, Indore Method, Vermi composting.



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3. Unit IV Techniques for Disposal of Solid Waste

4. Solid waste management techniques – solid waste management hierarchy, waste prevention
5. and waste reduction techniques
6. Land filling technique, Factors to be considered for site selection, Land filling methods-Area
7. method, Trench method and Ramp method, Leachate and its control, Biogas from landfill,
8. Advantages and disadvantages of landfill method, Recycling of municipal solid waste
9. Incineration of waste: Introduction of incineration process, Types of incinerators - Flash,
10. Multiple chamber Incinerators, Products of incineration process with their use, Pyrolysis of
11. waste – Definition, Methods
12. 113 Civil Engineering Curriculum Structure

Unit– V Biomedical and E-waste management

- Definition of Bio medical Waste.
- Sources and generation of Biomedical Waste and its classification
- Bio medical waste Management technologies.
- Definition, varieties and ill effects of E- waste,
- Recycling and disposal of E- waste.

Suggested learning resources:

- Gupta O.P, Elements of Solid Hazardous Waste Management, Khanna Book Publishing Co., Delhi Ed. 2018
- Bhide, A. D., Solid Waste Management, Indian National Scientific Documentation Centre, New Delhi.
- George Techobanoglous, Kreith, Frank., Solid Waste, McGraw Hill Publication, New Delhi.
- Sasikumar, K., Solid Waste Management, PHI learning, Delhi.
- Hosetti, B.B., Prospect and Perspectives of Solid Waste Management, New Age International Publisher.

Course outcomes:

After completing this course, student will be able to:

- Identify the sources of solid waste.
- Select the relevant method of collection and transportation of solid waste.
- Suggest an action plan for composting of solid waste.
- Devise suitable disposal technique for solid waste



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEO512
Course Title	Web Technology
Credits	3 (L: 3, T:0 P: 0)

Course Objectives: Giving the students the insights of the Internet programming and how to design and implement complete applications over the web.

Unit-I

Internet Fundamentals :Motivation for internetworking History and scope of internet Internet protocol and standardization 1.4 Role of ISP & Factors for choosing an ISP 1.5 Internet Service providers in India 1.6 Types of connectivity such as Dial Up, Leased, VSAT etc. 1.7 Internet server and client modules on various operating systems

Unit-II

TCP/ IP : 2.1 TCP/IP internet layering model 2.2 Reliable stream transport service(TCP) 2.3 Need for stream delivery 2.4 Properties of reliable delivery service 2.5 Providing reliability 2.6 Idea behind slide windows 2.7 Ports connection and end points Segment, stream, sequence number 2.8 TCP segment format,TCP header.

Unit-III

Internet Application and Services : 3.1 Email 3.2 Email networks 3.3 Email protocols 3.4 Format of an email address 3.5 Email routing 3.6 Email clients, POP3, IMAP 3.7 FTP 3.8 Public domain software 3.9 Types of FTP servers 3.10 FTP clients 3.11 Telnet 3.12 Telnet protocols 3.13 Server domain 3.14 Telnet clients 3.15 Terminal emulation 3.16 Internet Relay Chat 3.17 IRC network and servers 3.18 Channels

Unit-IV

E-Commerce : 4.1 Introduction to Electronic commerce 4.2 Modes of electronic commerce 4.3 Electronic data interchange 4.4 Migration to OPEN EDI 4.5 Electronic commerce with www/Internet 4.6 Different types of Electronics Payment System Credit card, Debit card, Smart Card, E-Cash ,E-Wallet

Unit-V

Web Publishing and Browsing : 5.1 Overview, SGML,HTML 5.2 Web hosting 5.3 CGL, Documents Interchange Standards 5.4 Components of Web Publishing, Document management 5.5 Web Page Design, Consideration and Principles 5.6 Search and Meta Search Engines 5.7



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WWW, Browser, HTTP, Publishing Tools

Unit-VI

Interactivity Tools : CGI, XML, ActiveX, VB Script, JAVA Script, Front Page, Adobe Dreamweaver, Flash

References

Internet working with TCP/IP VOL-1: Principles Protocol and Architecture by-Douglas EComer
–PHI Internet working with TCP/IP VOL-2: Design, Implementation and Internals by- Douglas E Comer, David L. Stevens- PHI

1. HTML: the Definitive guide –lby Chuck Musciano& Bui Kennedy4
2. Learning PHP, MySQL & JavaScript with jQuery, CSS & HTML5 by Robin Nixon

Course Outcome:

The student will be able to:

- Analyze a web page and identify its elements and attributes.
- Web pages using XHTML and Cascading Style Sheets.
- Build dynamic web pages using JavaScript (Client side programming).
- Create XML documents and Schemas.



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(Diploma EEE)

Credit System and Marks Distribution: -

Semester-VI												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	PCC	DEEP601	Power Electronics	3	0	0	3	30	70	21	100	35
2	PCC	DEEP602	Control Engineering	3	0		3	30	70	21	100	35
3	Huminitie s & social Sc. Course	DEEH603	Entrepreneurship	3	0	0	3	30	70	21	100	35
4	Mandator y Course	DEEM604	Environmental Science	0	0	0	0	30	70	21	100	35
5	OEC	DEEO(E)	Open Elective-II	3	0	0	3	30	70	21	100	35
6	OEC	DEEO(E)	Open Elective-III	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	PCC	DEEP651	Power Electronics Lab	0	0	2	1				50	25
2	PCC	DEEP652	Control Engineering Lab	0	0	2	1				50	25
3	PROJ	DE654	Major Project	0	0	8	4				50	25
4	SEM	DE655	Seminar	0	0	2	1				50	25
TOTAL							22					



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Elective Papers for semester-VI

Open Elective-II (Choose any one)												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	OEC	DEEO604(E)	PLC & SCADA	3	0	0	3	30	70	21	100	35
2	OEC	DEEO605(E)	Electrical hybrid Vehicle	3	0	0	3	30	70	21	100	35
3	OEC	DEEO606(E)	Biomass and Micro-hydro Power Plants	3	0	0	3	30	70	21	100	35
Open Elective-III (Choose any one)												
1	OEC	DEEO607(E)	Robotics	3	0	0	3	30	70	21	100	35
2	OEC	DEEO608(E)	Green Building & Energy Conservation	3	0	0	3	30	70	21	100	35
3	OEC	DEEO609(E)	Introduction to Artificial Intelligence	3	0	0	3	30	70	21	100	35



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma Electrical and Electronics Engineering

Semester – VI

Course Code	DEEP601
Course Title	Power Electronics
Credits	3(L:3 T: 0 P: 0)

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

Maintain the proper functioning of power electronic devices.

Unit-I

Power semiconductor devices PNP diodes, DIACS Thyristors, TRIACS, G.T.O. devices. Power Transistors, Power MOSFET, Rating, Losses and Cooling. Triggering circuits for SCR's, UJT, Blocking Oscillators, Schmitt trigger circuits – Power MOS gate drive circuits.

Unit-II

Uncontrolled and controlled Rectifiers : Single phase and poly phase Bridge rectifiers. Transformer ratings. Inductive load, free wheeling diodes. Converter operation: Overlap, power factor, inversion, regulation, P-pulse converters, power factor control via PWM converters

Unit-III

D.C. line commutation : Series and parallel capacitor turn off, resonant turn off, impulse commutation. D.C. Choppers : Principles, classification, use.

Unit-IV

Frequency conversion : Cycloconverter single and three phase circuits, blocked group operation, circulating current mode. Single phase and three phase inverters, constant voltage source and constant current source inverters, HF inverters for heating.

Unit-V

Application: D.C. and A.C. drives, S.M.P.S., Resonant converters, A.C. Line Filters, ratio, interference suppression. HDVC transmission.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING (Diploma EEE)

References:

1. Ramamoorthy M., An Introduction to Thyristors and their applications, East-West Press Pvt. Ltd., New Delhi, ISBN: 8185336679.
2. Sugandhi, Rajendra Kumar and Sugandhi, Krishna Kumar, Thyristors: Theory and Applications, New Age International (P) ltd. Publishers, New Delhi, ISBN: 978-0-85226-852-0.
3. Bhattacharya, S.K., Fundamentals of Power Electronics, Vikas Publishing House Pvt. Ltd. Noida. ISBN: 978-8125918530.
4. Jain & Alok , Power Electronics and its Applications, Penram International Publishing (India) Pvt. Ltd, Mumbai, ISBN: 978-8187972228.
5. Rashid , Muhammad, Power Electronics Circuits Devices and Applications, Pearson Education India, Noida, ISBN: 978-0133125900.

Course Outcome

At the end of the course, a student will be able to: Relate basic semiconductor physics to properties of power devices, and combine circuit mathematics and characteristics of linear and non-linear devices.



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(Diploma EEE)

Semester - VI

Course Code	DEEP651
Course Title	Power Electronics Lab
Credits	1(L:0 T: 0 P: 2)

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

Maintain the proper functioning of power electronic devices.

List of Experiment:

1. Test the proper functioning of power transistor.
2. Test the proper functioning of IGBT.
3. Test the proper functioning of DIAC to determine the break over voltage.
4. Determine the latching current and holding current using V-I characteristics of SCR.
5. Test the variation of R, C in R and RC triggering circuits on firing angle of SCR.
6. Test the effect of variation of R, C in UJT triggering technique.
7. Perform the operation of Class – A, B, C, turn off circuits.
8. Perform the operation of Class –D, E, F turn off circuits.
9. Use CRO to observe the output waveform of half wave-controlled rectifier with resistive load and determine the load voltage.
10. Draw the output waveform of Full wave-controlled rectifier with R load, RL load, free wheel diode and determine the load voltage.

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

- a) Select power electronic devices for specific applications.
- b) Maintain the performance of Thyristors.



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- c) Troubleshoot turn-on and turn-off circuits of Thyristors.
- d) Maintain phase controlled rectifiers.
- e) Maintain industrial control circuits



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Diploma Electrical and Electronics Engineering

Semester – VI

Course Code	DEEP602
Course Title	Control Engineering
Credits	3(L:3 T: 0 P: 0)

Course Objectives:

1. Demonstrate an understanding of the fundamentals of control systems.
2. Express and solve system equations in state-variable form (state variable models).
3. Determine the time and frequency-domain responses of first and second-order systems to step and sinusoidal (and to some extent, ramp) inputs.
4. Determine the (absolute) stability of a closed-loop control system

Unit I :

Introduction to Control Systems: Classification of control systems, Examples of Feedback Control Systems, Properties of Control Systems: Stability, steady-state & transient errors, Errors and Error constants, System types. Time response of system: Transient & steady state response of second order system and ramp response of second order system. Control actions: Proportional, integral,

Unit II:

Concept of state, state variable, state model. State variable formulation of control system, Relating transfer function with state model. Time response of state model of linear time-invariant system.

Unit III:

Introduction to Industrial Process Control. Modeling of Industrial process Control dynamics. Process Control terminology. Single loop control of standard first order process plants. Controller Implementation : Electronic, Analog,

Unit IV:

Pneumatic Controllers: P, P-I, P-D, P-I-D control, Controller tuning, Ziegler-Nichol's method. Examples and features of Level Control, Flow Control, Temperature Control, Feed-forward



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control, Ratio Control, Multi-loop and Cascade control System. Introduction to Programmable Logic Control (PLC), its Features & Architecture, Ladder Diagram and Simple case Studies.

Suggested Reading:

1. I.J. Nagrath, M. Gopal, Control Systems Engineering, New Age International Publishers
2. S. Hasan Saeed, Control Systems, Katson Books

Course Outcomes:

Upon Completing the Course, Student will able to:

- Understand the concept of LTI control systems, Importance of feedback in CS and stability concept
- Able to Design a Stable Control System
- Understand the difference between Linear and Digital Control Systems.



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Semester – VI

Course Code	DEEP652
Course Title	Control Engineering Lab
Credits	1(L:0 T: 0 P: 2)

Course Objectives:

- Will have a strong knowledge on MATLAB software.
- To study the concept of time response and frequency response of the system
- Students get the basic knowledge on practical control system applications on machines & electronic devices.
- This course aims to familiarize with the modeling of dynamical systems, to simulate and analyze the stability of the system using MATLAB.

List of Experiments (Expandable)

1. To generate the transfer function of open loop and closed loop transfer function of 1st order system in MATLAB.
2. To generate the transfer function of open loop and closed loop transfer function of 2nd order system in MATLAB.
3. Obtain creates a state-space model object representing the continuous-time state-space model with matrix A, B, C, D.
4. Conversion of the state space model to the transfer function model and then transfer function to state space model.
5. Obtain a pole –zero plot of a given transfer function.
6. Design a PD, PI controller in time and frequency domain.
7. Design a PID controller design in time and frequency domain.

Course Outcomes: At the end of the course, a student will be able to:

1. Discuss the need of software tools (MATLAB, PSPICE) to illustrate modeling and simulation of any system.
2. Classify and evaluate the performance parameters of a system and then with simulation prepare an advance tool to modify the values of the parameter of the system in order to



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meet the desired need.

3. Prepare professionals in laboratory to compute or to predict the characteristics of a system by visualizing experimental data and its graphical representation.
4. Evaluate possible causes of discrepancy in practical experimental observations in comparison to theory by introducing the concepts of different stability theorems



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Semester – V

Course Code	DEEH603
Course Title	Entrepreneurship
Credits	3 (L: 3, T:0 P: 0)

Course Objectives:

1. Acquiring Entrepreneurial spirit and resourcefulness.
2. Familiarization with various uses of human resource for earning dignified means of living.
3. Understanding the concept and process of entrepreneurship - its contribution and role in the growth and development of individual and the nation.
4. Acquiring entrepreneurial quality, competency, and motivation.
5. Learning the process and skills of creation and management of entrepreneurial venture.

UNIT-I

Introduction to Entrepreneurship Meaning and concept of entrepreneurship, the history of entrepreneurship development, role of entrepreneurship in economic development, Myths about entrepreneurs, agencies in entrepreneurship management and future of entrepreneurship types of entrepreneurs.

UNIT-II

The Entrepreneur Why to become entrepreneur, the skills/ traits required to be an entrepreneur, Creative and Design Thinking, the entrepreneurial decision process, skill gap analysis, and role models, mentors and support system, entrepreneurial success stories.

UNIT-III

E-Cell Meaning and concept of E-cells, advantages to join E-cell, significance of E-cell, various activities conducted by E-cell

UNIT-IV

Communication Importance of communication, barriers and gateways to communication, listening to people, the power of talk, personal selling, risk taking & resilience, negotiation.



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UNIT-V

Introduction to various form of business organization (sole proprietorship, partnership, corporations, Limited Liability company), mission, vision and strategy formulation

Course Outcome:

Upon completion of the course, the student will be able to demonstrate knowledge of the following topics:

1. Understanding the dynamic role of entrepreneurship and small businesses
2. Organizing and Managing a Small Business
3. Financial Planning and Control
4. Forms of Ownership for Small Business
5. Strategic Marketing Planning
6. New Product or Service Development
7. Business Plan Creation

Reference Books:

1. Udyamita by Dr.MMP. Akhourian and S.P Mishra, pub. By National Institute for Entrepreneurship and Small Business Development(NIESBUD),NSIC-PATC Campus, Okhla
2. Everyday Entrepreneurs-The harbingers of Prosperity and creators of Jobs-Dr.Aruna Bhargava.

SUGGESTED LEARNING RESOURCES:

S. No.	Title of Book	Author	Publication
1.	The Startup Owner's Manual: The Step-by-Step Guide for Building a Great Company	Steve Blank and Bob Dorf	K & S Ranch ISBN – 978-0984999392
2.	The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses	Eric Ries	Penguin UK ISBN – 978-0670921607
3.	Demand: Creating What People Love Before They Know They Want It	Adrian J. Slywotzky with Karl Weber	Headline Book Publishing ISBN – 978-0755388974



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4.	The Innovator's Dilemma: The Revolutionary Book That Will Change the Way You Do Business	Clayton M. Christensen	Harvard business ISBN: 978-142219602
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SUGGESTED SOFTWARE/LEARNING WEBSITES:

- <https://www.fundable.com/learn/resources/guides/startup>
- <https://corporatefinanceinstitute.com/resources/knowledge/finance/corporate-structure/>
- <https://www.finder.com/small-business-finance-tips>
- <https://www.profitbooks.net/funding-options-to-raise-startup-capital-for-your-business/>



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Diploma EEE Semester – V

Course Code	DEEM604
Course Title	Environmental Science
Credits	3 (L: 3, T:0 P: 0)

Course Objectives:

According to UNESCO (1971), the objectives of environmental studies are:

- (a) Creating the awareness about environmental problems among people.
- (b) Imparting basic knowledge about the environment and its allied problems.
- (c) Developing an attitude of concern for the environment.

Module-I

Introduction to environmental studies and Ecosystems

Multidisciplinary nature of environmental studies; Scope and importance; Concept of sustainability and sustainable development, ecosystem, Structure and function of ecosystem; Energy flow in an ecosystem: food chains, food webs and ecological succession, Case studies of the following ecosystems: Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries).

Module-II

Natural Resources: Renewable and Non-renewable Resources

Land resources and land use change; Land degradation, soil erosion and desertification, Deforestation: Causes and impacts due to mining, dam building on environment, forests, biodiversity and tribal populations, Water: Use and over-exploitation of surface and ground water, floods, droughts, conflicts over water (international & inter-state), Energy resources: Renewable and non-renewable energy sources, use of alternate energy sources, growing energy needs, case studies.

Module-III

Biodiversity and Conservation & Environmental Pollution

Levels of biological diversity: genetic, species and ecosystem diversity; Biogeographic zones of India; Biodiversity patterns and global biodiversity hot spots, India as a mega-biodiversity nation; Endangered and endemic species of India, Threats to biodiversity: Habitat loss, poaching of wildlife, man-wildlife conflicts, biological invasions; Conservation of biodiversity: In-situ and



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Ex-situ conservation , Environmental pollution: types, causes, effects and controls; Air, water, soil and noise pollution ,Nuclear hazards and human health risks ,Solid waste management: Control measures of urban and industrial waste ,Pollution case studies.

Module-IV

Environmental Policies- Practices and Human communities

Climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture, Environment Laws: Environment Protection Act; Air (Prevention & Control of Pollution) Act; Water (Prevention and control of Pollution) Act; Wildlife Protection Act; Forest Conservation Act. International agreements: Montreal and Kyoto protocols and Convention on Biological Diversity (CBD), Human population growth: Impacts on environment, human health and welfare, Resettlement and rehabilitation of project affected persons; case studies, Disaster management: floods, earthquake, cyclones and landslides. Case studies (e.g., factory pollution in Ranchi)

Reference Books

1. Blackwell's Concise Encyclopedia of Ecology by Peter Calow. ...
2. Blackwell's Concise Encyclopedia of Environmental Management by Peter Calow. ...
3. Conservation and Environmentalism by Robert C. ...
4. Encyclopedia of Biodiversity by Simon A. ...
5. Encyclopedia of Disasters by Angus M.

Course Outcomes

The course will empower the undergraduate students by helping them to:

- i. Gain in-depth knowledge on natural processes that sustain life, and govern economy.
- ii. Predict the consequences of human actions on the web of life, global economy and quality of human life.
- iii. Develop critical thinking for shaping strategies (scientific, social, economic and legal) for environmental protection and conservation of biodiversity, social equity and sustainable development.
- iv. Acquire values and attitudes towards understanding complex environmental-economic social challenges, and participating actively in solving current environmental problems and preventing the future ones.
- v. Adopt sustainability as a practice in life, society and industry



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Semester-VI

Course Code	DEEO604
Course Title	PLC & SCADA
Credits	3 (L: 3, T:0 P: 0)

COURSE OBJECTIVES:

1. To get familiar with industrial automation with PLC
2. To get familiar with industrial automation with SCADA
3. Knowing the basics of HMI

UNIT 1

Programmable Logic Controllers: Introduction, Parts of a PLC, Principles of Operation, Modifying the Operation, PLCs versus Computers, PLC Size and Application. PLC Hardware Components: The I/O Section, Discrete I/O Modules, Analog I/O Modules, Special I/O Modules, I/O Specifications, The Central Processing Unit (CPU), Memory Design, Memory Types, Programming Terminal Devices, Recording and Retrieving Data, Human Machine Interfaces (HMIs). Basics of PLC Programming: Processor Memory Organization, Program Scan, PLC Programming Languages, Relay-Type Instructions, Instruction Addressing, Branch Instructions, Internal Relay Instructions, Programming Examine If Closed and Examine If Open Instructions, Entering the Ladder Diagram, Modes of operation

UNIT 2

Developing Fundamental PLC Wiring Diagrams and Ladder Logic Programs: Electromagnetic Control Relays, Contactors, Motor Starters, Manually Operated Switches, Mechanically Operated Switches, Sensors, Output Control Devices, Seal-in Circuits, Latching Relays, Converting Relay Schematics into PLC Ladder Programs, Writing a Ladder Logic Program Directly from a Narrative Description. Programming Timers: Mechanical Timing Relays, Timer Instructions, On-Delay Timer Instruction, Off-Delay Timer Instruction, Retentive Timer, Cascading Timers.

UNIT 3

SCADA Fundamentals: Introduction, Open system: Need and advantages, Building blocks of SCADA systems, Remote terminal unit (RTU): Evolution of RTUs, Components of RTU, Communication subsystem, Logic subsystem, Termination subsystem



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UNIT 4

Human-Machine Interface (HMI): HMI components, HMI software functionalities, Situational awareness, Intelligent alarm filtering: Need and technique, Alarm suppression techniques, Operator needs and requirements,

UNIT 5

SCADA Systems: Building the SCADA systems, legacy, hybrid, and new systems, Classification of SCADA systems, SCADA implementation: A laboratory model: The SCADA laboratory, System hardware, System software, SCADA lab field design.

Text Books:

1. Programmable Logic Controllers Frank D Petruzella McGraw Hill 4th Edition, 2011
2. Power System SCADA and Smart Grids Mini S. Thomas CRC Press 3rd Edition, 2015

Course Outcomes: students will be able to

1. Do the ladder logic program in PLC
2. Understand the working with PLC and SCADA
3. Analyze the working of HMI



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(Diploma EEE)

Semester-VI

Course Code	DEEO605
Course Title	Electric Hybrid Vehicles
Credits	3 (L: 3, T:0 P: 0)

Course Objective:

1. Electric and Hybrid vehicles are gaining popularity globally.
2. This course introduces the fundamental concepts of electric, hybrid and autonomous vehicles, drive trains, electrical machines used, energy storage devices, charging systems and different communication protocols.

Module 1: Introduction

Conventional Vehicles: Basics of vehicle performance, vehicle power source characterization, transmission characteristics, mathematical models to describe vehicle performance.

Module 2: Introduction to Hybrid Electric Vehicles:

History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies.

Hybrid Electric Drive-trains: Basic concept of hybrid traction, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies, fuel efficiency analysis.

Module 3: Electric Trains

Electric Drive-trains: Basic concept of electric traction, introduction to various electric drive-train topologies, power flow control in electric drive-train topologies, fuel efficiency analysis. Electric Propulsion unit: Introduction to electric components used in hybrid and electric vehicles, Configuration and control of DC Motor drives, Configuration and control of Induction Motor drives, configuration and control of Permanent Magnet Motor drives, Configuration and control of Switch Reluctance Motor drives, drive system efficiency.

Module 4: Energy Storage Energy Storage:

Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles, Battery based energy storage and its analysis, Fuel Cell based energy storage and its analysis, Super Capacitor based energy storage and its analysis, Flywheel based energy storage and its analysis, Hybridization of different energy storage devices. Sizing the drive system: Matching the electric



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machine and the internal combustion engine (ICE), Sizing the propulsion motor, sizing the power electronics, selecting the energy storage technology, Communications, supporting subsystems

Module 5: Energy Management Strategies

Energy Management Strategies: Introduction to energy management strategies used in hybrid and electric vehicles, classification of different energy management strategies, comparison of different energy management strategies, implementation issues of energy management strategies. Case Studies: Design of a Hybrid Electric Vehicle (HEV), Design of a Battery Electric Vehicle (BEV).

Text / References:

1. C. Mi, M. A. Masrur and D. W. Gao, “ Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives”, John Wiley & Sons, 2011.
2. S. Onori, L. Serrao and G. Rizzoni, “ Hybrid Electric Vehicles: Energy Management Strategies” , Springer, 2015.
3. M. Ehsani, Y. Gao, S. E. Gay and A. Emadi, “Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design”, CRC Press, 2004.
4. T. Denton, “Electric and Hybrid Vehicles”, Routledge, 2016.

Course Outcomes

At the end of this course, students will demonstrate the ability to

- Understand the models to describe hybrid vehicles and their performance.
- Understand the different possible ways of energy storage.
- Understand the different strategies related to energy storage systems.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Semester-VI

Course Code	DEEO606
Course Title	Biomass And Micro-Hydro Power Plants
Credits	3 (L: 3, T:0 P: 0)

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

Maintain the efficient operation of various types of Biomass and Micro hydro power plants.

Unit– I Basics of Biomass-based Power Plants

Properties of solid fuel for biomass power plants: bagasse, wood chips, rice husk, municipal waste ,Properties of liquid and gaseous fuel for biomass power plants: Jatropha, bio-diesel gobar gas Layout of a Bio-chemical based (e.g. biogas) power plant:,Layout of a Thermo-chemical based (e.g. Municipal waste) power plant Layout of a Agro-chemical based (e.g. bio-diesel) power plant ,Selection of biomass power plants.

Unit– II Biomass Gasification Power Plants

The basic principle to convert Agriculture and forestry products and wood processing re- mains (including rick husks, wood powder, branches, offcuts, corn straws, rice straws, wheat straws, cotton straws, fruit shells, coconut shells, palm shells, bagasse, corncobs) into com- bustible gas ,General Construction and working of a typical gasifier

Power generating in gas engine:, Strengths and limitations of Agriculture and forestry products gasifier ,Preventive maintenance steps different types of biomass gasifiers.

Unit– III Different Types of Gasifiers

Construction and working of the following types of gasifiers:Rice Husk Gasification Power Plant and their specifications Straw Gasification Power Plant and their specifications Coconut shell, coconut peat, coconut husk, Gasification Power Plant and their specifications Bagasse/Sugar Cane Trash Gasification Power Plant and their specifications Gobar gas plant and its specifications Breakdown maintenance of biomass power plant at the module level.



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Unit– IV Micro-Hydro Power Plants

Locations of micro hydro power plant, Energy conversion process of hydro power plant.

Classification of hydro power plant: High, medium and low head., General Layouts of typical micro-hydro power plant. Strengths and limitations of micro hydro power plants

Unit– V Different types of Microhydropower plants

Construction and working of High head – Pelton turbine and their specifications Construction and working of Medium head – Francis turbine and their specifications Construction and working of Low head – Kaplan turbine and their specifications Preventive and breakdown maintenance of microhydro power plants , Safe Practices for microhydro power plants.

References:

1. Khoiyangbam, R S Navindu; Gupta and Sushil Kumar; Biogas Technology :Towards Sustain-able Development; TERI, New Delhi; ISBN: 9788179934043
2. David M. Buchla; Thomas E. Kissell; Thomas L. Floyd - Renewable Energy Systems, Pearson Education New Delhi , ISBN: 9789332586826,
3. Kothari, D.P. et al: Renewable Energy Sources and Emerging Technologies, PHI
4. Rachel, Sthuthi, Earnest, Joshua; -Wind Power Technologies, PHI Learning
5. O.P. Gupta, Energy Technology, Khanna Publishing House, ISBN: 978-93-86173-683

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

- a) Select the relevant biomass power plant
- b) Undertake the preventive maintenance of different types of biomass gasifiers
- c) Undertake the breakdown maintenance of different types of biomass gasifiers
- d) Maintain the optimized working of large wind power plants
- e) Maintain the optimized working of small wind turbines.
- f) Maintain the optimized working of micro hydro power plants.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Semester-VI

Course Code	DEEO607
Course Title	Robotics
Credits	3 (L: 3, T:0 P: 0)

Course Objectives:

- To introduce the basic concepts, parts of robots and types of robots.
- To make the student familiar with the various drive systems for robot, sensors and their applications in robots and programming of robots.
- To select the robots according to its usage.

Course Content:

UNIT-I: Fundamentals of Robotics: Introduction; Definition; Robot anatomy (parts) and its working; Robot Components: Manipulator, End effectors; Construction of links, Types of joints; Classification of robots; Cartesian, Cylindrical, Spherical, Scara, Vertical articulated; Structural Characteristics of robots; Mechanical rigidity; Effects of structure on control work envelope and work Volume; Robotwork Volumes, comparison; Advantages and disadvantages of robots.

Unit-II: Robotic Drive System and Controller: Actuators; Hydraulic, Pneumatic and Electrical drives; Linear actuator; Rotary drives; AC servo motor; DC servo motors and Stepper motors; Con- version between linear and rotary motion; Feedback devices; Potentiometers; Optical encoders; DC tachometers; Robot controller; Level of Controller; Open loop and Closed loop controller; Microprocessor based control system; Robot path control: Point to point, Continuous path control and Sensorbased path control; Controller programming.

Unit-III: Sensors: Requirements of a sensor; Principles and Applications of the following types of sensors: Position sensors (Encoders, Resolvers, Piezo Electric); Range sensors (Triangulation Principle, Structured lighting approach); Proximity sensing; Force and torque sensing.

Reference Books:

1. Introduction to Robotics: Analysis, Systems, Applications – Saeed B. Niku, Pearson Education Inc. New Delhi 2006.
2. Industrial Robotics: Technology, Programming and Applications – M.P. Groover, Tata McGraw Hill Co, 2001.



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3. Robotics Control, Sensing, Vision and Intelligence – Fu.K.S. Gonzalz.R.C and Lee C.S.G, Mc-Graw Hill Book Co, 1987.
4. Robotics for Engineers – Yoram Koren, McGraw Hill Book Co, 1992.
5. A Text book on Industrial Robotics – Ganesh S. Hedge, Laxmi Publications Pvt. Ltd., New Del-hi, 2008.

Course outcomes:

At the end of the course, the student will be able to:

1. Explain the robot anatomy, classification, characteristics of robot, advantages and disadvantages.
2. Explain the various robotic actuators on hydraulic, pneumatic and electrical drives.
3. Explain about various types of sensors and concepts on robot vision system.
4. Explain the various applications of robots



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(Diploma EEE)

Semester-VI

Course Code	DEEO608
Course Title	Biomass And Micro-Hydro Power Plants
Credits	3 (L: 3, T:0 P: 0)

Course Objectives:

Following are the objectives of this course:

- To know various aspects of green buildings
- To use different steps involved in measuring environmental impact assessment.
- To relate the construction of green building with prevailing energy conservation policy and regulations.
- To know and identify different green building construction materials.
- To learn different rating systems and their criteria.

Course Content:

Unit I : Introduction to Green Building and Design Features

- Definition of Green Building, Benefits of Green building, Components/features of Green Building, Site selection, Energy Efficiency, Water efficiency, Material Efficiency, Indoor Air Quality.
- Site selection strategies, Landscaping, building form, orientation, building envelope and fenestration,
- Material and construction techniques, roofs, walls, fenestration and shaded finishes,
- Advanced passive heating and cooling techniques, waste reduction during construction

Unit–II Energy Audit and Environmental Impact Assessment (EIA)

- Energy Audit: Meaning, Necessity, Procedures, Types, Energy Management Programs
- Environmental Impact Assessment (EIA): Introduction, EIA regulations, Steps in environmental impact assessment process, Benefits of EIA, Limitations of EIA, Environmental clearance for the civil engineering projects.

Unit– III Energy and Energy conservation

- Renewable Energy Resources: Solar Energy, Wind Energy, Ocean Energy, Hydro Energy, Biomass Energy



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- Non-renewable Energy Resources: Coal, Petroleum, Natural Gas, Nuclear Energy, Chemical
- Sources of Energy, Fuel Cells, Hydrogen, Biofuels.
- Energy conservation: Introduction, Specific objectives, present scenario, Need of energy conservation, LEED India Rating System and Energy Efficiency.

Unit– IV Green Building

- Introduction: Definition of Green building, Benefits of Green building,
- Principles: Principles and planning of Green building
- Features: Salient features of Green Building, Environmental design (ED) strategies for building construction.
- Process: Improvement in environmental quality in civil structure
- Civil Engineering Curriculum Structure 118
- Materials: Green building materials and products- Bamboo, Rice husk ash concrete, plastic bricks, Bagasse particle board, Insulated concrete forms. reuse of waste material-Plastic, rubber, Newspaper wood, Nontoxic paint, Green roofing

Unit V Rating System

- Introduction to (LEED) criteria,
- Indian Green Building council (IGBC) Green rating,
- Green Rating for Integrated Habitat Assessment. (GRIHA) criteria
- Heating Ventilation Air Conditioning (HVAC) unit in green Building
- Functions of Government organization working for Energy conservation and Audit (ECA)-
- National Productivity council (NPC)
- Ministry of New and Renewable Energy (MNRE)
- Bureau of Energy efficiency (BEE)

Suggested learning resources:

1. Kibert, C.J., Sustainable construction: Green Building design and Delivery, John Wiley Hoboken, New Jersey.
2. Chauhan, D S Sreevasthava, S K., Non-conventional Energy Resources, New Age International Publishers, New Delhi.
3. O.P. Gupta, Energy Technology, Khanna Publishing House, New Delhi
4. Jagadeesh, K S, Reddy Venkatta Rama & Nanjunda Rao, K S., Alternative Building Materials and Technologies, New Age International Publishers, Delhi.
5. Sam Kubba., Handbook of Green Building Design and Construction, Butterworth-Heinemann.
6. Means R S, Green Building - Project Planning and Cost Estimating, John Wiley & Sons



RKDF UNIVERSITY RANCHI

DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

7. Sharma K V, Venkateshaiah P., Energy Management and Conservation, IK International.

Course outcomes:

After completing this course, student will be able to:

- Identify various requirements for green building.
- Use different steps in environmental impact assessment.
- Relate the construction of green building with prevailing energy conservation policy and regulations.
- Supervise the construction of green building construction using green materials.
- Focus on criteria related to particular rating system for assessment of particular Green



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Semester - VI

Course Code	DEEO609
Course Title	Introduction to Artificial Intelligence
Credits	3 (L: 3, T:0 P: 0)

Course objective:

- To design a system capable of thinking for itself just like humans do. Currently, general AI is still under research, and efforts are being made to develop machines that have enhanced cognitive capabilities.

Unit-I

Introduction: Overview of Artificial intelligence- Problems of AI, AI technique, Tic - Tac - Toe problem. Intelligent Agents: Agents & environment, nature of environment, structure of agents, goal based agents.

Unit-II

Search techniques: Computer Science & Engineering Syllabus Solving problems by searching : problem solving agents, searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. Heuristic search strategies: Greedy best-first search, A* search, memory bounded heuristic search

Unit-III

Knowledge & reasoning: Knowledge representation issues, representation & mapping, approaches to knowledge representation, issues in knowledge representation. Using predicate logic

Unit-IV

Planning: Overview, components of a planning system, Goal stack planning, Hierarchical planning, other planning techniques. Natural Language processing : Introduction, Syntactic processing, semantic analysis, discourse & pragmatic processing.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING (Diploma EEE)

Unit-V

Learning: Forms of learning, inductive learning, learning decision trees, explanation based learning.

Text Books:

- Artificial Intelligence, Ritch & Knight, TMH
- Artificial Intelligence A Modern Approach, Stuart Russel Peter Norvig Pearson
- Introduction to Artificial Intelligence & Expert Systems, Patterson, PHI
- Poole, Computational Intelligence, OUP
- Logic & Prolog Programming, Saroj Kaushik, New Age International
- Expert Systems, Giarranto, VIKAS
- Artificial Intelligence, Russel, Pearson
- Identify problems where artificial intelligence techniques are applicable.
- Apply selected basic AI techniques; judge applicability of more advanced techniques.
- Participate in the design of systems that act intelligently and learn from experience.

Course Outcomes:

- Design user interfaces to improve human–AI interaction and real-time decision-making.
- Evaluate the advantages, disadvantages, challenges, and ramifications of human–AI augmentation.
- Design and develop symbiotic human–AI systems that balance the information processing power of computational systems with human intelligence and decision making.
- Explain the benefits, limitations, and tradeoffs of designing engaging and ethical conversational user interactions, including those supported by chatbots, smart speakers, and other AI-driven, voice-based technologies.
- Design and evaluate conversational interfaces for different users and contexts of use.



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DIPLOMA IN ELECTRICAL & ELECTRONICS ENGINEERING

(Diploma EEE)

Semester – VI

Course Code	DE654
Course Title	Major Project
Credits	4(L:3 T: 0 P: 8)

Course Objective:

This course is aimed to provide more weightage for project work. The project work could be done in the form of a summer project or internship in the industry or even a minor practical project in the college. Participation in any technical event/ competition to fabricate and demonstrate an innovative machine or product could be encouraged under this course.
